Design and Implementation of a Real-Time Impedance Controller for a 7-DOF Collaborative Robot

Entwurf und Implementierung eines Echtzeit-Impedanzreglers für einen 7-achsigen kollaborativen Roboter

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Declaration of Authorship

I hereby declare that this master's thesis entitled

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Munich, 30.09.2026 Heykel Khadhraoui

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ORANGE The argument and presentation are sufficient for the present thesis. Minor polishing may still be possible.

YELLOW The content is relevant, but the explanation is dense, repetitive, weakly supported, or harder to follow than necessary.

RED The passage retains a generic, templated, or textbook-restatement pattern that can read as machine-produced. This is a stylistic assessment, not a claim about authorship or an AI-detector result.

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Orange means sufficient, not perfect. A grey assessment does not make the reported data invalid; it identifies where the current evidence does not fully support the surrounding interpretation. Red is reserved for prose structure rather than technical correctness. Where a section contains both sound work and one important weakness, the more serious colour is used and the box names the exact concern.

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An unresolved concern can be raised beside the text as

The operator-controlled hold maintains the pose reached by the tool until continuation is confirmed. **comment:** “verify whether the same timing rule applies to the later hold”

A comment is a remark about the text, not a part of it. It appears only in this annotated copy and never in the submitted thesis.

Blue is used directly on new words or sentences and ends in a raised +. The two tints differ in hue so that the boundary survives greyscale printing.

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Abstract

ORANGE — sufficient. The abstract states the problem, implemented mechanism, principal measured ordering, null-space result, and scope limits without overloading the reader with the full parameter matrix.

Angular misalignment between a grinding tool and a surface can lead to edge-first contact and increased contact loads. In applications where the surface pose and tool mounting are subject to tolerances, the direction and magnitude of this misalignment are generally unknown before contact. This thesis investigates a Cartesian impedance control approach that uses the contact interaction itself to support the rotational alignment of the tool with a planar surface.

A real-time Cartesian impedance controller was implemented for a seven-degree-of-freedom (DOF) Franka Emika Panda. Translational and rotational compliance are defined relative to the surface geometry. The controller allows the virtual centre of compliance to be shifted away from the tool centre point (TCP). This shift introduces translation–rotation coupling into the Cartesian impedance, so that the commanded normal press can generate an additional moment that supports the required alignment rotation without prescribing an additional rotational trajectory.

The experimental investigation examined the influence of directional Cartesian stiffness and of the tangential position of the virtual compliance centre. The reported angular conditions and responses are referenced to the configured surface geometry; the physical surface normal and instantaneous physical tool–surface angle were not measured independently. The compliance-centre position produced the largest response variation over the tested parameter ranges. An assisting tangential displacement lies perpendicular to the required rotation direction. Reversing the sign of the initial angular deviation requires the displacement to be placed on the opposite side of the TCP, while changing the required tangent-plane rotation direction requires a corresponding change in the displacement di-

rection.

Within the tested conditions, every non-zero tangential displacement that assisted alignment was therefore specific to the initial angular-deviation direction and sign. The TCP-centred condition introduces no preferred tangential direction and is the direction-independent fixed centre of compliance among the tested positions for the investigated contact configuration. A displaced centre can instead be selected when the required alignment direction is known and additional rotational alignment authority is needed.

A complementary null-space study showed that projected damping reduced redundant joint motion. Singular-value conditioning was active while the commanded disturbance was applied and strongly suppressed the resulting net redundant displacement under the tested disturbance direction. This response therefore represents disturbance rejection during its application rather than a subsequent return of the redundant configuration.

Kurzfassung

ORANGE — sufficient. The German summary remains aligned with the English abstract in structure, certainty, findings, and limitations.

Eine Winkelabweichung zwischen einem Schleifwerkzeug und einer Fläche kann zu Kantenkontakt und erhöhten Kontaktbelastungen führen. Bei Anwendungen, in denen die Lage der Fläche und die Werkzeugaufnahme Toleranzen aufweisen, sind Richtung und Größe dieser Winkelabweichung vor dem Kontakt in der Regel nicht bekannt. Diese Arbeit untersucht einen kartesischen Impedanzregelungsansatz, bei dem die Kontaktinteraktion selbst die rotatorische Ausrichtung des Werkzeugs an einer ebenen Fläche unterstützt.

Für einen 7-achsigen Franka Emika Panda wurde ein kartesischer Echtzeit-Impedanzregler implementiert. Die translatorische und rotatorische Nachgiebigkeit wird relativ zur Flächengeometrie definiert. Das virtuelle Nachgiebigkeitszentrum kann gegenüber dem Werkzeugbezugspunkt (TCP) verschoben werden. Dadurch entsteht eine Kopplung zwischen Translation und Rotation innerhalb der kartesischen Impedanz, sodass die kommandierte normale Anpressbewegung ein zusätzliches Moment erzeugen kann, das die erforderliche Ausrichtungsdrehung unterstützt, ohne eine zusätzliche rotatorische Trajektorie vorzugeben.

Experimentell wurden der Einfluss der richtungsabhängigen kartesischen Steifigkeit sowie der tangentialen Lage des virtuellen Nachgiebigkeitszentrums untersucht. Die berichteten Winkelbedingungen und Reaktionen beziehen sich auf die konfigurierte Flächengeometrie; die physische Flächennormale und der momentane physische Winkel zwischen Werkzeug und Fläche wurden nicht unabhängig gemessen. Innerhalb der geprüften Parameterbereiche führte die Lage des Nachgiebigkeitszentrums zur größten Variation der Ausrichtungsreaktion. Eine unterstützende tangentiale Verschiebung liegt senkrecht zur erforderlichen Drehrichtung. Wird das Vorzeichen der anfänglichen Winkelabweichung umgekehrt, muss

die Verschiebung auf die gegenüberliegende Seite des TCP gelegt werden. Ändert sich die erforderliche Drehrichtung in der Tangentialebene, muss sich entsprechend auch die Richtung der Verschiebung ändern.

Unter den untersuchten Bedingungen war damit jede nicht verschwindende tangentiale Verschiebung, die die Ausrichtung unterstützte, von Richtung und Vorzeichen der anfänglichen Winkelabweichung abhängig. Die TCP-zentrierte Einstellung gibt keine bevorzugte tangentiale Richtung vor und stellt unter den getesteten Lagen das richtungsunabhängige feste Nachgiebigkeitszentrum für die untersuchte Kontaktkonfiguration dar. Ein verschobenes Nachgiebigkeitszentrum kann dagegen gezielt eingesetzt werden, wenn die erforderliche Ausrichtungsrichtung bekannt ist und zusätzliche rotatorische Ausrichtungswirkung benötigt wird.

Eine ergänzende Nullraumuntersuchung zeigte, dass die projizierte Dämpfung redundante Gelenkbewegungen reduzierte. Die Singularwert-Konditionierung war während der kommandierten Störung aktiv und unterdrückte unter der untersuchten Störungsrichtung die daraus resultierende Netto-Verschiebung im Nullraum deutlich. Dieses Verhalten beschreibt somit eine Unterdrückung der Störwirkung während ihrer Einwirkung und keine nachträgliche Rückführung der redundanten Konfiguration.

Contents

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List of Abbreviations

ORANGE — sufficient. The abbreviations are separated from the symbol list and their printed forms are controlled centrally.

CoC	Centre of Compliance
CSV	Comma-Separated Values
DH	Denavit–Hartenberg
DOF	Degree of Freedom
EE	End-Effector
E-stop	Emergency Stop
FCI	Franka Control Interface
FCU	Franka Control Unit
F/T	Force/Torque
IP	Internet Protocol
LDLT	Lower–Diagonal–Lower-Transpose Factorisation
OS	Operating System
PREEMPT_RT	Real-Time Preemption Patch
ROS	Robot Operating System
SE(3)	Special Euclidean Group in three dimensions
SO(3)	Special Orthogonal Group in three dimensions
SVD	Singular Value Decomposition
TCP	Tool Centre Point

List of Symbols

ORANGE — sufficient. The notation is extensive but well organised, and the unit column makes the mixed translational and rotational quantities traceable.

Frames and Kinematics

$\{0\}$	Robot base frame	[-]
$\{\mathrm{EE}\}$	End-effector frame used by the robot model	[-]
$\{d\}$	Desired end-effector and TCP reference frame	[-]
$\{S\}$	Surface frame, with axes t_1, t_2 and n_s	[-]
n	Number of joints ($n = 7$)	[-]
$q, \ \dot{q}$	Joint position / velocity vector	[rad], [rad/s]
p	Position / translation vector	[m]
R	Rotation matrix	[-]
T	Homogeneous transformation matrix	[-]
$J(q)$	Geometric Jacobian; linear rows above angular rows	[m/rad], [-]

Pose and Error

$p_{\mathrm{EE}},\ p_{\mathrm{TCP}}$	End-effector / tool-centre-point position; $p_{\mathrm{TCP}} = p_{\mathrm{EE}}$ in the implemented model	[m]
$R_{\mathrm{EE}},\ T_{\mathrm{EE}}$	End-effector orientation / transformation	[-]
$R_{\mathrm{EE,clearance}},\ p_{\mathrm{TCP,clearance}}$	End-effector orientation captured at the clearance transition and held as the contact-establishment orientation reference / TCP position captured at the same instant	[-], [m]
$R_{\mathrm{CE,start}},\ R_{\mathrm{CE,end}}$	Measured end-effector orientations at the beginning / end of contact establishment	[-]
R_{CE}	Current-to-reference relative rotation over contact establishment, $R_{\mathrm{CE,start}}R_{\mathrm{CE,end}}^{\top}$	[-]
$u_{\mathrm{CE}},\ \phi_{\mathrm{CE}}$	Axis / angle of R_{CE}	[-], [rad]
$p_d,\ R_d,\ T_d$	Desired position / orientation / transformation	[m], [-]
ΔR_0	Current-to-desired relative rotation in the base frame, $R_d R_{\mathrm{EE}}^{\top}$	[-]
ϕ	Orientation-error angle	[rad]
$u,\ [u]_\times$	Base-frame orientation-error axis / its skew matrix	[-]
e_p	Translational (position) error	[m]
e_R	Base-frame rotational error used by the command	[rad]
$\dot{x}$	Cartesian velocity (twist)	[m/s], [rad/s]

Cartesian Stiffness and Damping

$K_p,\ D_p$	Generic translational stiffness / damping block; a frame or state index is added when needed	[N/m], [N s/m]
$K_R,\ D_R$	Generic rotational stiffness / damping block; a frame or state index is added when needed	[N m/rad], [N m s/rad]
$K_{p,t_1},\ K_{p,t_2},\ K_{p,n}$	Translational stiffness entries along t_1, t_2, and n_s	[N/m]
$K_{R,t_1},\ K_{R,t_2},\ K_{R,n}$	Rotational stiffness entries about t_1, t_2, and n_s	[N m/rad]
$D_{p,t_1},\ D_{p,t_2},\ D_{p,n}$	Translational damping entries along t_1, t_2, and n_s	[N s/m]
$D_{R,t_1},\ D_{R,t_2},\ D_{R,n}$	Rotational damping entries about t_1, t_2, and n_s	[N m s/rad]
$K_0,\ D_0$	Complete base-frame Cartesian stiffness / damping, assembled block diagonal from the rotated translational and rotational blocks	[N/m], [N m/rad]; [N s/m], [N m s/rad]

Dynamics and Torques

$M(q)$	Joint-space inertia matrix	$[\mathrm{kg\,m^2}]$
$\Lambda_0(q),\ \Lambda_S(q)$	Regularised operational-space inertia in base coordinates / resolved along the surface-frame directions	Translational: [kg]; rotational: $[\mathrm{kg\,m^2}]$
T_R	Block rotation $\mathrm{diag}(R_{\mathrm{surface}}, R_{\mathrm{surface}})$	[-]
$\lambda_i(q)$	Directional Cartesian inertia, the ith diagonal entry of $\Lambda_S(q)$	Translational: [kg]; rotational: $[\mathrm{kg\,m^2}]$
$k_i,\ d_i$	Directional stiffness / inertia-scaled damping coefficient	Translational: [N/m], [N s/m]; rotational: [N m/rad], [N m s/rad]
$\zeta,\ \varepsilon$	Selected damping factor / operational-space-inertia regularisation constant	[-], representation-dependent
F	Cartesian wrench	[N], [N m]
$f,\ m$	Complete commanded Cartesian force / moment at the TCP	[N], [N m]
$f_n,\ f_t$	Normal / tangential component of the complete commanded force, $f_n = F_n n_s$ and $f_t = f - f_n$	[N], [N]
$K,\ D$	Full Cartesian stiffness / damping matrices	K: [N/m], [N m/rad]; D: [N s/m], [N m s/rad]
τ	Joint torque vector	[N m]
$\tau_c,\ \tau_g$	Coriolis / gravity compensation torque	[N m]
τ_{cart}	Cartesian impedance torque, $J^\top(q)F$	[N m]
τ_{cmd}	Complete commanded joint torque	[N m]

Surface and Contact Geometry

n_s	Configured surface-reference normal copied into the controller and used in the evaluation	[-]
n_{phys}	Physical surface normal; not measured independently in the reported experiments	[-]
$t_1,\ t_2$	Surface tangent directions (unit)	[-]
R_{surface}	Surface-frame orientation $[\,t_1\ \ t_2\ \ n_s\,]$ in the base frame	[-]
$n_{\mathrm{Tool,EE}},\ n_{\mathrm{Tool,0}},\ n_d$	Tool-face normal expressed in the EE frame, fixed by calibration / the same direction expressed in the base frame, $n_{\mathrm{Tool,0}} = R_{\mathrm{EE}} n_{\mathrm{Tool,EE}}$ / desired tool-normal direction used by the controller	[-]
$\theta_{\mathrm{cmd}},\ u_{\mathrm{cmd}},\ R_{\mathrm{cmd}}$	Commanded tool-orientation offset relative to the configured surface reference, $\theta_{\mathrm{cmd}} = \theta_{t_1} t_1 + \theta_{t_2} t_2$ / its unit rotation axis / the rotation matrix it produces through $R(u, \phi)$	[rad], [-], [-]
$\theta_a,\ \theta_{a,t_1},\ \theta_{a,t_2}$	Generic tangent-plane angular deviation used to derive the compliance-centre direction rule / its surface-tangent components	[rad]
p_s	Point on the configured surface	[m]
$e_{x_0},\ e_{y_0},\ e_{z_0}$	Unit vectors along the base-frame axes x_0, y_0 and z_0	[-]
p_c	Centre of compliance	[m]

r_c	TCP-to-compliance-centre displacement, $p_c - p_{\mathrm{TCP}}$, as the point-shift transformation and the experimental plots and tables use it; where no frame index is shown, the base-frame representation is implied	[m]
$r_{c,0}$, $r_{c,S}$, $r_{c,\mathrm{EE}}$	Representations of the same displacement in the base, surface and end-effector frames, with $r_{c,0} = R_{\mathrm{surface}} r_{c,S} = R_{\mathrm{EE}} r_{c,\mathrm{EE}}$	[m]
r_{c,t_1}, r_{c,t_2}, $r_{c,n}$	Surface-frame components of r_c along t_1, t_2 and n_s	[m]
$r_{\mathrm{Tool,EE}}$, p_{Tool}	Selected tool-point offset from the TCP, expressed in the EE frame / selected tool-point position in the base frame	[m]
r_{Tool}	Selected tool-point offset from the TCP, $p_{\mathrm{Tool}} - p_{\mathrm{TCP}}$	[m]
$p_{\mathrm{Tool,clearance}}$, $p_{\mathrm{Tool},0}$, $p_{\mathrm{Tool},d}$	Selected tool point at the clearance transition / its projection onto the configured surface reference / its desired contact-establishment position	[m]
s_{app}, v_{app}, $s_{\mathrm{app,max}}$	Commanded approach travel of the TCP along $-n_s$ / its configured rate / its configured maximum	[m], [m/s], [m]
$p_{\mathrm{TCP,start}}$, $t_{\mathrm{app},0}$, $t_{\mathrm{app,max}}$	TCP position captured when surface approach begins / that instant / the instant at which s_{app} reaches $s_{\mathrm{app,max}}$	[m], [s], [s]
h_i	Signed clearance of tool-face corner i from the configured surface reference	[m]

h_{Tool}	Minimum clearance of the rectangular tool face from the configured surface reference, $h_{\text{Tool}} = \min_i h_i$	[m]
$h_{\text{clearance}}$	Configured tool clearance at which the transition from approach to contact establishment occurs	[m]
ε_{sel}	Tool-point selection tolerance	[m]
s_{CE}, v_{CE}, $s_{\text{CE,max}}$	Commanded contact-establishment press coordinate of the selected tool point along $-n_s$ / its configured rate / its configured endpoint	[m], [m/s], [m]
$t_{\text{CE,0}}$, $t_{\text{CE,max}}$	Instant of the clearance transition / instant at which s_{CE} reaches $s_{\text{CE,max}}$	[s], [s]
$t_{\text{CE,start}}$, $t_{\text{CE,end}}$	Measurement instants at the beginning / end of Contact Establishment	[s]
m_R	Rotational-impedance contribution to the commanded moment, $K_R e_R - D_R \omega_{\text{EE}}$, present at every compliance-centre position	[N m]

Compliance and Alignment

$\mathrm{Ad}(r_c)$	TCP-to-compliance-centre point-shift adjoint	[-]
K_c, D_c	Block-diagonal stiffness / damping at the centre of compliance	K: [N/m], [N m/rad]; D: [N s/m], [N m s/rad]
$F_c,\ F_{\mathrm{TCP}}$	Cartesian wrench at the centre of compliance / corresponding wrench at the TCP	[N], [N m]
$K_{\mathrm{TCP}}, D_{\mathrm{TCP}}$	Centre-of-compliance stiffness and damping mapped by congruence to the TCP	K: [N/m], [N m/rad]; D: [N s/m], [N m s/rad]
$\theta_{t_1},\ \theta_{t_2}$	Surface-frame components of the commanded tool orientation offset	[rad]
$\theta_{0,S}$	Pose-based initial angular-deviation vector in surface coordinates	[rad]
$\theta_{0,t_1},\ \theta_{0,t_2}$	Achieved pose-based initial angular deviation about the surface tangents, relative to the configured reference	[rad]
$\gamma_0,\ \gamma_S$	Signed contact-establishment response vector in base / surface coordinates: the end-of-contact-establishment orientation change relative to the held contact-establishment reference	[rad], [rad]
$\gamma_{t_1},\ \gamma_{t_2},\ \gamma_n$	Surface-frame components of the signed contact-establishment response; a component reducing an initial deviation carries the same sign as that deviation	[rad]

$r_{c,t}$	Tangential part of the compliance-centre displacement, $r_{c,t_1}t_1 + r_{c,t_2}t_2$; its magnitude is written $\|r_{c,t}\|$	[m]
$F_n,\ M_{t_i}$	Signed commanded normal force / commanded TCP moment about surface tangent t_i	[N], [N m]

Null-Space and Singular Value Decomposition

J^+	Thresholded-SVD Moore–Penrose pseudoinverse	[rad/m], [-]
$N_q,\ N_\tau$	Joint-velocity / torque null-space projector	[-]
$\sigma_i,\ \sigma_{\min}$	Jacobian singular value / σ_6 conditioning indicator. The geometric Jacobian mixes linear and angular rows, so a singular value carries no single physical unit and its numerical value depends on the chosen Jacobian representation and length unit; it is used as a relative indicator under one fixed convention	[-]
v_7	Structural null direction of the seven-joint Jacobian	[-]
$\dot{q}_{\text{null}}$	Projected joint velocity in the numerical null space	[rad/s]
d_{null}	Null-space damping	[N m s/rad]
k_σ	Sigma-conditioning torque magnitude	[N m]
α_{probe}	Null-direction sampling angle	[rad]

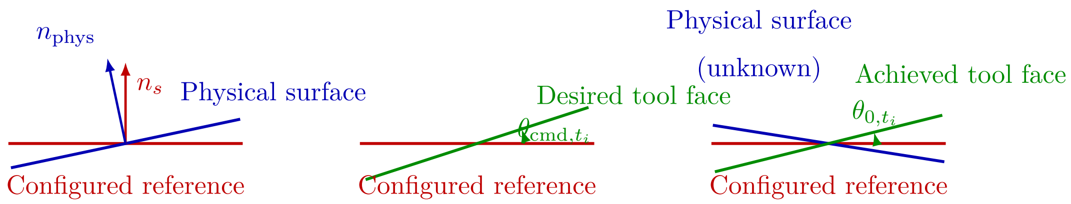
$\Delta\sigma_{\mathrm{min,dist}}$	Experimental change in $\sigma_{\min}$ over the disturbance interval, $\sigma_{\min}(9\,\mathrm{s}) - \sigma_{\min}(5\,\mathrm{s})$; distinct from the controller's Δ_σ	[-]
$\tau_d,\ \tau_\sigma,\ \tau_{\mathrm{null}}$	Projected damping / sigma / total null-space torque	[N m]
E_N	Cumulative projected null-space motion; integrated projected joint-velocity magnitude during the disturbance interval, converted to degrees for reporting	[°]
v_{ref}	Fixed redundant comparison direction, taken from the condition without null-space torque	[-]
$\Delta q_{\mathrm{null}},\ \Delta\eta_{\mathrm{dist}}$	Net projected joint displacement over the disturbance interval / its signed component along v_{ref}	[rad]

Chapter 1

Introduction

1.1 Motivation

Industrial manipulators commonly use position control to track commanded trajectories. In the controller, the surface is represented by a configured reference with normal n_s. The physical surface normal n_{phys} can differ from this reference because of placement and calibration tolerances. Tool mounting adds a further source of tool–surface angular uncertainty. Consequently, commanding the tool face parallel to the configured surface does not guarantee parallelism with the physical surface. The remaining mismatch becomes apparent during contact establishment and can produce edge-first contact. Rigid position control resists the corresponding corrective motion and can generate high contact loads.



(a) Surface references (b) Commanded offset (c) Contact entry

Figure 1.1: Configured surface reference, commanded tool offset, and contact-entry geometry. The physical normal is unknown to the controller, and the achieved reference-relative angle need not equal the command.

The physical surface-reference difference was not measured independently during the reported campaign. Reproducible entry conditions were instead created by varying the desired tool orientation relative to the configured surface. The command θ_{cmd} specifies this deliberate reference-relative offset. The pose-based angle θ_0 describes the orientation actually reached at contact entry. These quantities do not represent a measured physical surface angle.+

Torque-controlled collaborative robots allow contact compliance to be specified directly and account for a growing share of industrial deployments [12, 24]. Joint torque sensing and a torque-level command interface determine how the robot yields under contact as well as where it moves. In surface finishing, this capability can reduce the loads caused by angular tool–surface misalignment. It can also permit the tool face to rotate towards the surface while maintaining the commanded normal press. The reported response is obtained from the end-effector rotation during Contact Establishment. The physical tool–surface angle was not measured independently.

Impedance control specifies this yielding behaviour as a dynamic relation between motion and interaction force [11]. Cartesian impedance control defines the relation for the end-effector pose. Its translational and rotational compliance can therefore be assigned independently along selected Cartesian directions. During edge-first contact, the wrench contains both force and moment components. Stiffness and damping defined from the contact geometry allow part of this moment to rotate the tool towards the surface. The stiffness and damping matrices consequently determine both the contact response and the free-space tracking behaviour.

1.2 Related Work

The relevant literature comprises three areas: Cartesian impedance control, real-time implementation on torque-controlled robots, and the placement of the centre of compliance (CoC). The first area is Cartesian impedance control itself. The controller uses standard serial-manipulator kinematics [21, 3, 17]: homogeneous transformations for representing the end-effector pose and the Jacobian for relating joint motion to Cartesian motion. Khatib established the operational-space formulation, in which a Cartesian wrench F is mapped to joint torques τ through $J^\top$ [13]; this mapping is adopted in the controller. Hogan defined the impedance relation between end-effector motion and interaction force [11]. Albu-Schäffer et al. implemented Cartesian impedance control using joint-torque sensing [1], which corresponds to the torque-controlled interface used in this thesis.

The second area concerns the real-time implementation of Cartesian impedance control on torque-controlled robots. Mayr et al. implemented a modular Cartesian impedance controller for the Franka Emika Panda that combines real-time model access, null-space control, signal treatment, and command limiting [15]. The controller developed in this thesis uses the libfranka model interface to obtain the required robot state and model quantities during the real-time control loop. It also includes a selectable null-space torque τ_{null} for influencing the redundant joint motion without changing the commanded Cartesian motion.

Recent work has applied compliant and force-controlled strategies directly to robotic surface finishing. Alt et al. combined perception, planning, and force-controlled execution for robotic surface treatment [2]. Min et al. integrated demonstrated motion and force skills with impedance control on a 7-DOF collaborative robot [16]. Wu et al. used adaptive variable impedance control to regulate the contact force during robotic grinding [27]. These studies address force-controlled surface treatment. The present work instead examines contact-induced angular alignment through directional Cartesian stiffness and a virtual CoC.

The third area concerns the location of the CoC. Fasse and Broenink formulated spatial impedance about a selected point, so that translation and rotation become coupled through the position of this point [5]. This formulation provides the theoretical basis for the compliance-centre shift used in this thesis. The same principle appears mechanically in remote-centre-compliance devices, in which elastic elements are arranged so that the tool behaves as if it rotates about a virtual point located away from the mechanism itself [18, 26]. More recent mechanically compliant and variable-compliance-centre strategies apply this principle to peg-in-hole insertion [25, 10]. In the mechanically compliant mechanisms, elastic bending produces the alignment motion rather than software control alone. In these approaches, the compliance is arranged so that contact forces reduce misalignment instead of increasing it. Most of these studies focus on insertion tasks and define the compliance centre through either the mechanical design or a strategy adapted to the hole geometry. In this thesis, an equivalent effect is created in software by shifting the virtual CoC away from the TCP. The independent effects of this position and the axis-specific stiffness during planar tool alignment are then investigated on a torque-controlled robot.

Suomalainen et al. experimentally investigated how compliance-centre placement and compliant-axis selection influence alignment in an assembly task [23]. Friedrich et al. subsequently proposed an active remote CoC that combines passive compliance with active force control [9]. These approaches further illustrate the role of compliance location in contact alignment. The present work examines a software-defined virtual compliance centre for planar tool–surface alignment.

1.3 Problem Statement

The angular relation between the tool and the surface plane is uncertain before contact because the surface placement and tool mounting both carry tolerances. The sign of the initial angle and its tangent-plane rotation direction are therefore unknown when a run starts. Under rigid position control, the contact geometry can generate high moments while the controller resists the corrective motion. In contrast, finite rotational compliance allows contact-induced rotation, whereas a displaced CoC adds a selectable coupling between translation and rotation.

The displacement r_c points from the TCP to the virtual CoC. Its tangential component enters the commanded wrench through a cross product with the translational force and thereby selects an additional rotational moment. The direction and sign of this moment depend on the displacement direction. A displaced centre chosen before contact therefore encodes a preferred alignment direction.

The investigation evaluates a fixed CoC for conditions in which the sign and tangent-plane direction of the initial angular misalignment are unknown. In this thesis, a direction-independent fixed centre denotes a centre position that can be selected without this prior information. The assessment is limited to the investigated contact configuration, parameter range, and tested centre positions. The term describes direction independence of the selected reference and carries no claim of global optimality.

Contact-induced alignment forms the Contact Establishment state of the intended surface-contact sequence. The tool approaches the surface with an angular offset, establishes contact, and rotates under the compliant interaction. Sustained tangential motion would follow after alignment. However, the reported measurements cover the Contact Establishment state. An operator-controlled hold then maintains the reached Cartesian reference before grinding.

The alignment response depends on the directional stiffness, contact geometry, compliance-centre displacement, robot configuration, and compliance of the tool mounting. The experiments therefore vary the selected controller parameters while keeping the calibrated geometry, the robot configuration, and the tool mounting unchanged. A commanded tool orientation offset sets the desired tool direction relative to the configured surface reference. The experiments are identified by the pose-based initial angular deviation reached at contact entry, because the finite orientation tolerance separates that angle from the nominal command. Chapters 2 and 3 introduce the frame conventions and controller formulation used in the experiments.

1.4 Scope and Contributions

The scope covers contact establishment against one configured surface reference. Sustained grinding and adaptive selection of the compliance-centre position are outside the experimental evaluation. The main contributions are the following.

- A real-time Cartesian impedance controller with surface-relative directional stiffness and damping and a shiftable virtual CoC was implemented on a Franka Emika Panda.
- The influence of directional stiffness and tangential compliance-centre position on contact-induced end-effector rotation was characterised experimentally.
- A projected null-space controller combining joint damping and singular-value conditioning was implemented and evaluated separately in Cartesian pose hold.

The contact study distinguishes a direction-independent fixed reference from a condition-dependent displaced centre. The former can be selected before the sign and tangent-plane direction of the initial angular deviation are known. The latter adds rotational authority through the commanded moment $r_c \times f$ when the required direction is available.

1.5 Thesis Structure

Chapter 2 establishes the impedance law, frame conventions, and compliance-centre shift. Their real-time implementation on the Franka Emika Panda is described in Chapter 3. The experimental method and results follow in Chapters 4 and 5. Chapter 6 states the resulting conclusions and future work.

Chapter 2

Theoretical Background

This chapter presents the mathematical basis of the Cartesian impedance controller. The required frame conventions and pose errors are introduced first. The subsequent sections derive the kinematic mappings, the impedance wrench F, the compliance-centre transformation, and the null-space torque τ_{null}.

2.1 Reference Frames

The following right-handed coordinate frames are used in the theoretical formulation and the controller [21, 3].

- **Base frame $\{0\}$:** The base frame is fixed to the robot mounting surface. Its origin is the robot base reference point used by the kinematic model. For the upright Panda configuration used in this thesis, the z_0-axis is normal to the mounting surface, while the x_0- and y_0-axes lie in the mounting plane. Unless stated otherwise, Cartesian positions, directions, velocities, forces, and moments are expressed in this frame.
- **Joint frames $\{1\}, \{2\}, \ldots, \{n\}$:** The joint frames follow the robot kinematic convention and are required for forward kinematics and Jacobian construction; the real-time controller obtains the corresponding quantities from the libfranka model.
- **End-effector frame $\{\mathrm{EE}\}$:** The end-effector frame is attached to the controlled TCP and moves with the tool. In the implemented robot model, the end-effector origin and TCP are identified, so $p_{\text{TCP}} = p_{\text{EE}}$. The physical tool face is represented separately by its calibrated geometric offset from this frame. The position p_{EE}

and orientation R_{EE}, both expressed relative to the base frame, define the current Cartesian pose of the robot. The axes x_{EE}, y_{EE}, and z_{EE} describe the current end-effector orientation. The configured tool approach axis is defined in this frame and is transformed to the base frame using R_{EE}.

- **Desired end-effector frame $\{d\}$:** The desired frame represents the commanded TCP pose. Its origin is p_d, and its orientation is R_d, both expressed relative to the base frame. The difference between the desired and current end-effector frames provides the translational and rotational errors used by the impedance controller.
- **Surface frame $\{S\}$:** The surface frame is tied to the configured surface rather than to the moving tool. Its origin is placed at the configured surface point p_s. Its orientation R_{surface} is fixed by the two surface tangents t_1 and t_2 and the surface normal n_s. Its construction is given in Section 2.5.
  The directions t_1 and t_2 lie in the surface plane and are mutually orthogonal. The unit normal n_s is defined to point outward from the surface towards the free-space region above the plate. This sign convention is kept throughout the contact sequence, so motion towards the surface is along $-n_s$. The surface frame is used to describe the commanded tool orientation offset and to assign stiffness and damping values according to the two tangential directions and the normal direction.

Figure 2.1 shows these frames together with the pose error between the current and the desired end-effector frame.

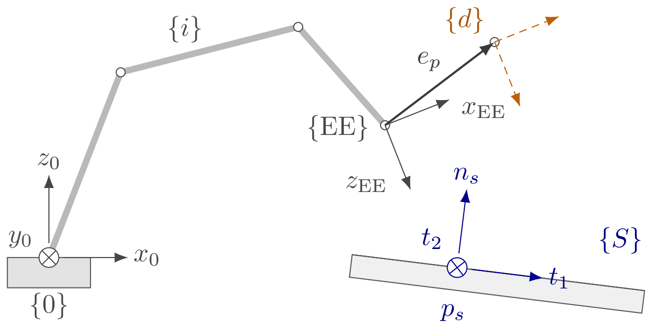


Figure 2.1: Frames used in the controller and the end-effector pose error.

2.2 Rigid-Body Pose and Pose Error

Unless stated otherwise, Cartesian quantities are expressed in the robot base frame. A Cartesian pose consists of a position $p \in \mathbb{R}^3$ and an orientation $R \in SO(3)$, collected in the homogeneous transformation T:

$$T = \begin{bmatrix} R & p \\ 0^\top & 1 \end{bmatrix}. \tag{2.1}$$

The measured end-effector pose is therefore

$$T_{\mathrm{EE}} = \begin{bmatrix} R_{\mathrm{EE}} & p_{\mathrm{EE}} \\ 0^\top & 1 \end{bmatrix}. \tag{2.2}$$

The controller obtains T_{EE} directly from the libfranka robot state. Rotation matrices are used throughout; the pose is not converted to Euler angles.

2.2.1 Cartesian Pose Error

The desired end-effector transformation T_d is

$$T_d = \begin{bmatrix} R_d & p_d \\ 0^\top & 1 \end{bmatrix}, \tag{2.3}$$

where p_d and R_d are the desired TCP position and orientation, respectively. The current end-effector pose is denoted by T_{EE}.

The relative pose-error transformation T_{err} maps the current end-effector frame to the desired frame:

$$T_{\mathrm{err}} = T_{\mathrm{EE}}^{-1} T_d = \begin{bmatrix} R_{\mathrm{EE}}^\top R_d & R_{\mathrm{EE}}^\top (p_d - p_{\mathrm{EE}}) \\ 0^\top & 1 \end{bmatrix}. \tag{2.4}$$

When the current and desired poses coincide, $T_{\mathrm{err}} = I_4$. The rotational and translational blocks of Equation (2.4) therefore both represent zero error in this case.

Position error. The translational block of Equation (2.4) expresses the position error e_p in the current end-effector frame. The position error e_p used by the Cartesian impedance law is expressed in the base frame:

$$e_p = p_d - p_{\mathrm{EE}} \in \mathbb{R}^3. \tag{2.5}$$

A positive position error e_p therefore points from the current TCP position towards the desired TCP position.

Orientation error. The rotational block of Equation (2.4) is expressed in the current end-effector frame. The equivalent relative rotation in the robot base frame is ΔR_0:

$$\Delta R_0 = R_d R_{\mathrm{EE}}^\top. \tag{2.6}$$

This expression follows by conjugating the current-frame block:

$$R_{\mathrm{EE}}(R_{\mathrm{EE}}^\top R_d)R_{\mathrm{EE}}^\top = R_d R_{\mathrm{EE}}^\top.$$

It represents the correction from the current orientation to the desired orientation directly in base coordinates. If both orientations coincide, then $\Delta R_0 = I$.

For use in the impedance controller, ΔR_0 is represented by a rotation angle ϕ and a unit rotation axis $u \in \mathbb{R}^3$ [21, 17]. The rotation-matrix and axis–angle representations can be converted in both directions. Given a unit axis u and an angle ϕ, the corresponding rotation matrix is obtained from Rodrigues' formula,

$$R(u, \phi) = I + \sin\phi\,[u]_\times + (1 - \cos\phi)[u]_\times^2. \tag{2.7}$$

For the relative rotation considered here, $R(u, \phi) = \Delta R_0$. The skew-symmetric matrix of the rotation axis is

$$[u]_\times = \begin{bmatrix} 0 & -u_z & u_y \\ u_z & 0 & -u_x \\ -u_y & u_x & 0 \end{bmatrix}. \tag{2.8}$$

Conversely, when the rotation matrix ΔR_0 is known, the angle follows from

$$\operatorname{tr}(\Delta R_0) = 1 + 2\cos\phi, \tag{2.9}$$

and therefore

$$\phi = \arccos\left(\frac{\operatorname{tr}(\Delta R_0) - 1}{2}\right), \qquad \phi \in [0, \pi]. \tag{2.10}$$

Subtracting the transpose of ΔR_0 isolates its skew-symmetric part:

$$\frac{1}{2}\left(\Delta R_0 - \Delta R_0^\top\right) = \sin\phi\,[u]_\times. \tag{2.11}$$

For $\sin\phi \neq 0$, the rotation axis is obtained from

$$u = \frac{1}{2\sin\phi}\begin{bmatrix}\Delta R_{0,32} - \Delta R_{0,23} \\ \Delta R_{0,13} - \Delta R_{0,31} \\ \Delta R_{0,21} - \Delta R_{0,12}\end{bmatrix}. \tag{2.12}$$

Multiplying the base-frame unit axis by the angle gives the controller orientation error e_R directly:

$$e_R = \phi u \in \mathbb{R}^3. \tag{2.13}$$

The direction of e_R defines the axis of the required correction, while its magnitude is the angular error in radians. When the current and desired orientations coincide, $\phi = 0$ and $e_R = 0$. Together, $e_p = p_d - p_{\text{EE}}$ and e_R are expressed in the base frame and point in the translational and rotational directions required to reduce the pose error.

The complete Cartesian error e stacks the position error e_p and orientation error e_R used by the impedance controller:

$$e = \begin{bmatrix} e_p \\ e_R \end{bmatrix} \in \mathbb{R}^6. \tag{2.14}$$

2.3 Kinematics of a 7-DOF Manipulator

Having defined the end-effector pose as the controlled variable, its relation to the joint configuration is introduced next.

For the joint configuration $q \in \mathbb{R}^7$, forward kinematics defines the end-effector pose $T_{\mathrm{EE}}(q)$. The Denavit–Hartenberg convention fixes the joint and frame indexing [4, 3], but the real-time controller obtains the pose and geometric Jacobian from the libfranka model interface.

The geometric Jacobian $J(q) \in \mathbb{R}^{6\times 7}$ maps joint velocity to the base-frame end-effector twist,

$$\begin{bmatrix} \dot{p}_{\mathrm{EE}} \\ \omega_{\mathrm{EE}} \end{bmatrix} = J(q)\dot{q}. \tag{2.15}$$

Its transpose maps the commanded Cartesian wrench to joint torque. These two mappings are the kinematic relations required by the impedance and null-space controllers.

2.4 Cartesian Impedance Control

The geometric Jacobian $J(q)$ is also used through its transpose to map a Cartesian wrench F to joint torques τ [13]. Before introducing this mapping, the desired Cartesian interaction behaviour is defined.

In this thesis, Cartesian impedance control is formulated as a spring–damper relation between the end-effector motion and the commanded wrench [11, 19, 15]. The Cartesian pose error e, defined in Section 2.2.1, produces a restoring wrench through the stiffness matrix K, while the Cartesian velocity $\dot{x}$ produces an opposing wrench through the damping matrix D. Their entries determine the translational and rotational compliance in the selected compliance directions [20]. Physically, the stiffness terms determine how strongly a pose error generates a restoring force or moment, while the damping terms oppose the corresponding Cartesian motion.

2.4.1 Impedance Wrench

The impedance wrench is evaluated in the robot base frame. The corresponding translational stiffness and damping matrices are $K_{p,0}, D_{p,0} \in \mathbb{R}^{3\times 3}$, while the rotational matrices are $K_{R,0}, D_{R,0} \in \mathbb{R}^{3\times 3}$. These matrices are not generally diagonal in the base frame because the direction-dependent stiffness and damping are defined relative to the surface frame. Their base-frame representation therefore changes with the surface-frame orientation, even when the local matrix entries remain unchanged.

The commanded Cartesian force f is given by the translational spring–damper law [11, 15]

$$f = K_{p,0} e_p + D_{p,0} \left(\dot{p}_d - \dot{p}_{\mathrm{EE}}\right) \in \mathbb{R}^3, \tag{2.16}$$

where $\dot{p}_d$ and $\dot{p}_{\mathrm{EE}}$ are the desired and measured linear velocities expressed in the base frame. For a fixed position setpoint, $\dot{p}_d = \mathbf{0}$, and the damping term becomes $-D_{p,0}\dot{p}_{\mathrm{EE}}$.

The commanded Cartesian moment m is

$$m = K_{R,0} e_R - D_{R,0} \omega_{\mathrm{EE}} \in \mathbb{R}^3, \tag{2.17}$$

where ω_{EE} is the measured end-effector angular velocity expressed in the base frame. Since no angular reference velocity is commanded, the rotational damping acts directly against ω_{EE}.

The Cartesian wrench F stacks the force and moment:

$$F = \begin{bmatrix} f \\ m \end{bmatrix} = \begin{bmatrix} K_{p,0} e_p + D_{p,0} \left(\dot{p}_d - \dot{p}_{\mathrm{EE}}\right) \\ K_{R,0} e_R - D_{R,0} \omega_{\mathrm{EE}} \end{bmatrix} \in \mathbb{R}^6. \tag{2.18}$$

The wrench is ordered as force followed by moment, and all its components are expressed in the base frame. It is mapped to joint torques τ_{cart} through the Jacobian transpose in the following subsection.

Equation (2.18) defines the impedance about the TCP, with translation and rotation decoupled. Section 2.7 extends the same law to an impedance defined about a displaced centre of compliance.

2.4.2 Joint-Torque Control Law

By the principle of virtual work, the commanded Cartesian wrench F is mapped to joint torques τ_{cart} through the transpose of the geometric Jacobian $J(q)$ [13]:

$$\tau_{\text{cart}} = J^{\top}(q)F, \tag{2.19}$$

where $\tau_{\text{cart}} \in \mathbb{R}^n$ is the Cartesian impedance torque contribution and $J^{\top}(q) \in \mathbb{R}^{n\times 6}$. Substituting Equation (2.18) gives

$$\tau_{\text{cart}} = J^{\top}(q) \begin{bmatrix} K_{p,0}e_p + D_{p,0}\left(\dot{p}_d - \dot{p}_{\text{EE}}\right) \\ K_{R,0}e_R - D_{R,0}\omega_{\text{EE}} \end{bmatrix}. \tag{2.20}$$

The stiffness and damping matrices in this expression are obtained from their surface-coordinate representations. A change in the surface-frame orientation therefore changes the Cartesian wrench F and the resulting joint-torque contribution τ_{cart}, even when the local stiffness and damping values remain unchanged.

2.4.3 Coriolis and Gravity Compensation in libfranka

For comparison, a general model-compensated torque balance τ_{model} combines Cartesian impedance, gravity, and Coriolis/centrifugal compensation:

$$\tau_{\text{model}} = \tau_{\text{cart}} + \tau_g(q) + \tau_c(q, \dot{q}), \tag{2.21}$$

where $\tau_g(q)$ is the gravity torque and $\tau_c(q, \dot{q})$ is the Coriolis and centrifugal torque.

These terms are handled differently by the Franka Control Interface. The Coriolis vector provided by the libfranka model is added explicitly to the command because it is not compensated internally. A separate gravity term is not added. In external joint-torque mode, the robot compensates gravity and motor friction internally, and the commanded torque is interpreted as excluding gravity [7]. Adding $\tau_g(q)$ would therefore compensate gravity twice.

The Cartesian-only implemented command $\tau_{\text{cmd,cart}}$ is consequently

$$\tau_{\text{cmd,cart}} = \tau_{\text{cart}} + \tau_c(q, \dot{q}) = J^\top(q)F + \tau_c(q, \dot{q}), \tag{2.22}$$

which follows the official libfranka Cartesian impedance example [8]. The null-space torque τ_{null} introduced later is added to this expression to obtain the complete commanded torque in Equation (2.68).

2.5 Surface Frame

The surface frame provides the tangential and normal directions used for the Cartesian controller quantities. Its orientation is constructed from the unit surface normal n_s, expressed in the robot base frame. The sign of n_s is chosen so that it points outward from the surface towards the free-space region above the plate, which makes $-n_s$ the approach and contact-establishment direction.

The first surface tangent is chosen to follow the base-frame x_0-direction as closely as possible while remaining in the surface plane. Since the surface is not exactly parallel to the base x_0y_0-plane, e_{x_0} is projected onto the surface plane and normalised:

$$t_1 = \frac{e_{x_0} - n_s\left(n_s^\top e_{x_0}\right)}{\|e_{x_0} - n_s\left(n_s^\top e_{x_0}\right)\|}. \tag{2.23}$$

The second tangent completes the right-handed frame,

$$t_2 = n_s \times t_1. \tag{2.24}$$

The construction requires e_{x_0} not to be parallel to n_s, and t_1 and t_2 are then orthogonal unit vectors in the surface plane. Because the plane and its normal do not depend on that reference direction, a different in-plane reference would rotate t_1 and t_2 within the surface without changing the surface geometry, so an axis-specific result holds with respect to the direction that was configured, the first-tangent hint recorded in Appendix C. The tangents obtained for the reported experiments are given in Section 4.2.

The orientation of the surface frame relative to the robot base frame is

$$R_{\text{surface}} = \begin{bmatrix} t_1 & t_2 & n_s \end{bmatrix}. \tag{2.25}$$

Surface-frame quantities therefore use the coordinate order

$$\begin{bmatrix} t_1 & t_2 & n_s \end{bmatrix}^\top.$$

This convention is used consistently for the directional Cartesian quantities. For example, K_{p,t_1}, K_{p,t_2}, and $K_{p,n}$ denote the translational stiffness entries along the two surface tangents and the surface normal, respectively. The same ordering is used for the rotational stiffness and damping and for surface-frame components of the compliance-centre displacement.

2.6 Stiffness and Damping Representation

The Cartesian impedance law of Section 2.4 is evaluated in the robot base frame. The directional stiffness and damping values are specified relative to the configured surface reference, because the relevant directions for the contact problem are the two surface tangents t_1 and t_2 and the surface normal n_s.

The translational and rotational stiffness and damping matrices are first defined in surface coordinates as

$$K_{p,S} = \begin{bmatrix} K_{p,t_1} & 0 & 0 \\ 0 & K_{p,t_2} & 0 \\ 0 & 0 & K_{p,n} \end{bmatrix}, \qquad D_{p,S} = \begin{bmatrix} D_{p,t_1} & 0 & 0 \\ 0 & D_{p,t_2} & 0 \\ 0 & 0 & D_{p,n} \end{bmatrix}, \tag{2.26}$$

$$K_{R,S} = \begin{bmatrix} K_{R,t_1} & 0 & 0 \\ 0 & K_{R,t_2} & 0 \\ 0 & 0 & K_{R,n} \end{bmatrix}, \qquad D_{R,S} = \begin{bmatrix} D_{R,t_1} & 0 & 0 \\ 0 & D_{R,t_2} & 0 \\ 0 & 0 & D_{R,n} \end{bmatrix}. \tag{2.27}$$

The common axis order is $[t_1, t_2, n_s]$.

For use in the base-frame impedance law, these matrices are rotated into the robot base frame:

$$K_{p,0} = R_{\text{surface}} K_{p,S} R_{\text{surface}}^{\top}, \qquad D_{p,0} = R_{\text{surface}} D_{p,S} R_{\text{surface}}^{\top}, \tag{2.28}$$

$$K_{R,0} = R_{\text{surface}} K_{R,S} R_{\text{surface}}^{\top}, \qquad D_{R,0} = R_{\text{surface}} D_{R,S} R_{\text{surface}}^{\top}. \tag{2.29}$$

The complete base-frame stiffness and damping matrices are then

$$K_0 = \begin{bmatrix} K_{p,0} & 0 \\ 0 & K_{R,0} \end{bmatrix}, \qquad D_0 = \begin{bmatrix} D_{p,0} & 0 \\ 0 & D_{R,0} \end{bmatrix}. \tag{2.30}$$

At this stage translation and rotation remain decoupled. The centre-of-compliance transformation introduced in Section 2.7 can subsequently introduce coupling between the two blocks.

The transformation changes only the coordinate representation of the stiffness and damping matrices. Their physical directions remain tied to the configured surface reference. For example, $K_{p,n}$ acts along n_s, even when the surface normal is not aligned with a robot-base axis.

2.6.1 Directional Cartesian Inertia and Inertia-Scaled Damping

The damping coefficients are calculated from the stiffness and the Cartesian inertia along the same direction. The regularised operational-space inertia $\Lambda_0(q)$ in base coordinates is

$$\Lambda_0(q) = \Big(J(q) M^{-1}(q) J^{\top}(q) + \varepsilon I_6\Big)^{-1}, \tag{2.31}$$

where $M(q)$ is the joint-space inertia and $\varepsilon > 0$ is a regularisation constant. The block rotation $T_R = \operatorname{diag}(R_{\text{surface}}, R_{\text{surface}})$ resolves this matrix along the surface-frame directions. The resulting surface-resolved operational-space inertia $\Lambda_S(q)$ and its directional entries $\lambda_i(q)$ are

$$\Lambda_S(q) = T_R^{\top} \Lambda_0(q) T_R, \qquad \lambda_i(q) = [\Lambda_S(q)]_{ii}\,. \tag{2.32}$$

The index i denotes one translational or rotational surface-frame direction. Each controller state or configuration supplies the active stiffness entry k_i for that direction. The corresponding inertia-scaled damping coefficient $d_i(q)$ is

$$d_i(q) = 2\zeta\sqrt{\lambda_i(q)k_i}, \tag{2.33}$$

where ζ is the selected damping factor. For one decoupled direction, $\zeta = 1$ gives the critical-damping coefficient of the corresponding local second-order model.

2.7 Centre of Compliance

The surface-frame transformations of Equations (2.28) and (2.29) change the directions along which the Cartesian stiffness and damping act, while leaving their reference point at the tool centre point (TCP). Independently of that orientation choice, the Cartesian impedance can be defined about a virtual CoC p_c. Shifting this point away from the TCP introduces translation–rotation coupling into the Cartesian impedance [17, 5, 22, 14]. The construction is applied during Contact Establishment, where it lets the normal pressing action generate a rotational moment as well as a translational contact load.

For a dual-arm assembly task, Suomalainen et al. experimentally showed that compliance-centre placement and compliant-axis selection affect the alignment response [23]. This experimental result motivates the placement study in this thesis. The point-shift adjoint below remains derived from the spatial-transformation formulation.

The displacement from the TCP to the CoC is

$$r_c = p_c - p_{\mathrm{TCP}} \in \mathbb{R}^3, \qquad p_{\mathrm{TCP}} = p_{\mathrm{EE}}. \tag{2.34}$$

The vector r_c describes only the virtual shift of the impedance reference point.

Let $[r_c]_\times$ denote the skew-symmetric matrix satisfying $[r_c]_\times a = r_c \times a$. Under the local small-angle formulation, the translational error referred to the CoC is

$$e_{p,c} = e_p - [r_c]_\times e_R. \tag{2.35}$$

The negative sign follows from the selected displacement convention. The vector from the

TCP to p_c is r_c itself, and therefore

$$e_R \times r_c = -[r_c]_\times e_R.$$

The stacked error at the CoC collects $e_{p,c}$ and the rotational error e_R. The point-shift adjoint $\mathrm{Ad}(r_c)$ maps the corresponding TCP error onto it:

$$e_c = \begin{bmatrix} e_{p,c} \\ e_R \end{bmatrix} = \mathrm{Ad}(r_c) \begin{bmatrix} e_p \\ e_R \end{bmatrix}, \qquad \mathrm{Ad}(r_c) = \begin{bmatrix} I_3 & -[r_c]_\times \\ 0 & I_3 \end{bmatrix}. \tag{2.36}$$

The surface-frame rotation and the point shift have different functions. The transformations in Equations (2.28) and (2.29) change the principal directions of the impedance. The point-shift adjoint $\mathrm{Ad}(r_c)$ changes the point about which the impedance is defined.

After the stiffness and damping have been expressed in the base frame, the translational and rotational stiffness and damping matrices are assembled at the CoC as

$$K_c = \begin{bmatrix} K_p & 0 \\ 0 & K_R \end{bmatrix}, \qquad D_c = \begin{bmatrix} D_p & 0 \\ 0 & D_R \end{bmatrix}. \tag{2.37}$$

All blocks in Equation (2.37) are expressed in the base-frame orientation. The compliance-centre wrench is $F_c = K_c e_c$, and transforming this power-conjugate wrench gives the corresponding TCP wrench $F_{\mathrm{TCP}} = \mathrm{Ad}(r_c)^\top F_c$. The matrices at the TCP therefore follow from the congruence transformations

$$K_{\mathrm{TCP}} = \mathrm{Ad}(r_c)^\top K_c \mathrm{Ad}(r_c) = \begin{bmatrix} K_p & -K_p[r_c]_\times \\ -[r_c]_\times^\top K_p & K_R + [r_c]_\times^\top K_p [r_c]_\times \end{bmatrix} \tag{2.38}$$

and

$$D_{\mathrm{TCP}} = \mathrm{Ad}(r_c)^\top D_c \mathrm{Ad}(r_c) = \begin{bmatrix} D_p & -D_p[r_c]_\times \\ -[r_c]_\times^\top D_p & D_R + [r_c]_\times^\top D_p [r_c]_\times \end{bmatrix}. \tag{2.39}$$

The off-diagonal blocks introduce translation–rotation coupling. A translational position or velocity error can therefore contribute to the commanded moment, while a rotational error or velocity can contribute to the commanded force. For $r_c = 0$ the point shift disappears and the block-diagonal impedance at the TCP is recovered.

2.7.1 Commanded Moment

The Cartesian wrench generated by the impedance controller is $F = [f^\top,\ m^\top]^\top$, where f and m denote the complete commanded force and moment at the TCP.

Without a CoC displacement, the commanded moment is given by the ordinary rotational impedance,

$$m_R = K_R e_R - D_R \omega_{\mathrm{EE}}. \tag{2.40}$$

A non-zero CoC displacement adds the translation–rotation coupling contribution $r_c \times f$, generated by the complete commanded force. The complete commanded TCP moment is therefore

$$m = m_R + r_c \times f. \tag{2.41}$$

This relation is exact for the point-shifted stiffness and damping matrices above. The controller evaluates the complete shifted spring–damper wrench and does not separately command an elastic or a damping coupling moment.

For $r_c = 0$ the coupling contribution vanishes and $m = m_R$. The TCP-centred condition is therefore a zero-coupling condition rather than a zero-moment condition. The rotational compliance remains finite and the commanded rotational impedance remains active.

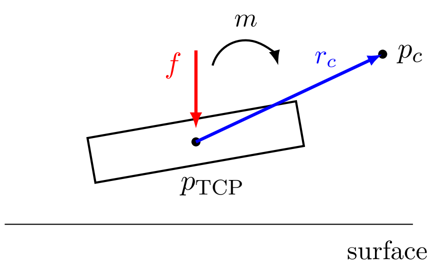


Figure 2.2: Centre-of-compliance displacement and commanded TCP moment. Shifting the virtual centre from the TCP by r_c introduces the coupling contribution $r_c \times f$.

2.7.2 Surface-Frame and Tangential Compliance-Centre Displacement

For surface contact the CoC displacement is resolved along the two surface tangents and the surface normal. Its coordinates in the surface frame are

$$r_{c,S} = \begin{bmatrix} r_{c,t_1} \\ r_{c,t_2} \\ r_{c,n} \end{bmatrix}. \tag{2.42}$$

Since $R_{\text{surface}} = [\, t_1 \;\; t_2 \;\; n_s \,]$ holds those surface axes expressed in the base frame, the same physical displacement in base coordinates is

$$r_{c,0} = R_{\text{surface}} r_{c,S} = r_{c,t_1} t_1 + r_{c,t_2} t_2 + r_{c,n} n_s, \tag{2.43}$$

with the inverse relation $r_{c,S} = R_{\text{surface}}^{\top} r_{c,0}$. The tangential part collects the two in-plane components,

$$r_{c,t} = r_{c,t_1} t_1 + r_{c,t_2} t_2, \tag{2.44}$$

while $r_{c,n} n_s$ is the part along the surface normal. The quantities have to be kept apart: r_{c,t_1} and r_{c,t_2} are signed scalar coordinates, $r_{c,t}$ is the tangential vector they form, and its magnitude is written $\|r_{c,t}\|$. For a purely tangential displacement $r_{c,n} = 0$, and therefore $\|r_c\| = \|r_{c,t}\|$.

A non-zero displacement is specified by the frame in which its components are held constant as well as by its magnitude and direction. The surface-fixed definition is given by Equation (2.43). For a tool-fixed definition, the end-effector orientation carries the configured vector into the base frame:

$$r_{c,0} = R_{\text{EE}} r_{c,\text{EE}}. \tag{2.45}$$

For a surface-fixed definition $r_{c,S}$ remains constant, and because the surface frame is fixed for one controller session the base-frame displacement is constant during contact establishment. For a tool-fixed definition $r_{c,\mathrm{EE}}$ remains constant instead, so $r_{c,0}$ rotates with the end effector. Section 3.2.5 describes both implementations.

The commanded force enters through $r_c \times f$, and it decomposes exactly into a normal and a tangential component,

$$f = f_n + f_t, \qquad f_n = F_n n_s, \qquad f_t = f - f_n, \tag{2.46}$$

with $F_n = n_s^\top f$ the signed commanded normal force. The normal component is the projection of f onto n_s, and the tangential component is what remains of f in the surface plane. The complete CoC contribution is therefore

$$r_c \times f = r_c \times f_n + r_c \times f_t. \tag{2.47}$$

Both terms remain part of the complete commanded moment m.

The normal-press term simplifies. Substituting $r_c = r_{c,t} + r_{c,n} n_s$ and using $f_n \parallel n_s$, the normal component of the displacement drops out, $r_{c,n} n_s \times f_n = \mathbf{0}$, which leaves

$$r_c \times f_n = r_{c,t} \times f_n. \tag{2.48}$$

Only the tangential part of the displacement produces a moment from the commanded normal press. A displacement along the surface normal contributes nothing to this term. Consequently, the moment generated by the normal press depends on the tangential projection $r_{c,t}$, not on the total distance $\|r_c\|$ between the TCP and the CoC.

The right-handed surface frame satisfies

$$t_1 \times t_2 = n_s, \qquad t_2 \times n_s = t_1, \qquad t_1 \times n_s = -t_2. \tag{2.49}$$

Consider a small angular deviation that the contact response should reduce. Its non-zero tangent-plane direction is written as

$$\theta_a = \theta_{a,t_1} t_1 + \theta_{a,t_2} t_2. \tag{2.50}$$

For a prescribed tangential magnitude $\|r_{c,t}\|$, the displacement used to reduce this deviation is selected perpendicular to its direction,

$$r_{c,t} = \|r_{c,t}\| \frac{\theta_{a,t_1} t_2 - \theta_{a,t_2} t_1}{\sqrt{\theta_{a,t_1}^2 + \theta_{a,t_2}^2}}. \tag{2.51}$$

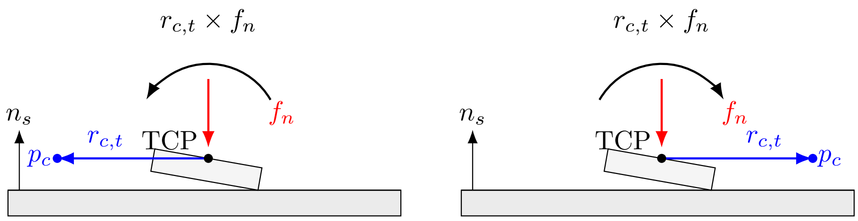
The choice fixes the direction; the magnitude remains a free parameter of the configuration. Using Equation (2.49), the normal-press contribution becomes

$$r_{c,t} \times f_n = \|r_{c,t}\| F_n \frac{\theta_{a,t_1} t_1 + \theta_{a,t_2} t_2}{\sqrt{\theta_{a,t_1}^2 + \theta_{a,t_2}^2}}. \tag{2.52}$$

During the contact-establishment press $F_n < 0$, so this contribution acts opposite to θ_a and therefore tends to reduce the deviation. For the two principal directions the rule reduces to

$$r_{c,t} = +\|r_{c,t}\| t_2 \quad \text{for} \quad +\theta_{a,t_1}, \qquad r_{c,t} = -\|r_{c,t}\| t_1 \quad \text{for} \quad +\theta_{a,t_2}, \tag{2.53}$$

giving $r_{c,t} \times f_n = \|r_{c,t}\| F_n t_1$ and $r_{c,t} \times f_n = \|r_{c,t}\| F_n t_2$ respectively. Reversing the angular deviation reverses $r_{c,t}$ and therefore reverses this contribution. Figure 2.3 shows the geometry for the two signs.



(a) selected displacement (b) reversed displacement

Figure 2.3: Normal-force contribution to the compliance-centre coupling moment for opposite tangential displacements: (a) the displacement selected from the angular-deviation direction; (b) the displacement with the opposite sign.

The complete commanded moment retains the form of Equation (2.41). The normal-force decomposition is used only to explain the directional effect of the CoC displacement. Tangential commanded-force components and damping effects remain contained in the complete wrench generated by the controller.

The coupling vanishes for $r_c = 0$. Conversely, if K_p is nonsingular, zero coupling implies $r_c = 0$. Since $\mathrm{Ad}(r_c)$ is invertible, the congruence transformation also preserves definiteness:

$$K_c \succeq 0 \implies K_{\mathrm{TCP}} \succeq 0, \qquad K_c \succ 0 \implies K_{\mathrm{TCP}} \succ 0. \tag{2.54}$$

The same statements apply to D_c and D_{TCP}. Negative off-diagonal entries therefore represent coupling terms rather than negative stiffness directions. Closed-loop robot–controller–contact behaviour with state switching was assessed experimentally under the operating conditions reported in Chapter 4.

This formulation is a local small-motion model in which the displacement is treated as fixed during one controller evaluation. A surface-fixed $r_{c,S}$ therefore keeps $r_{c,0}$ constant during contact establishment, whereas a tool-fixed displacement does not, and the controller updates the base-frame vector and the shifted stiffness and damping matrices during each control cycle in that case.

The virtual CoC p_c is consequently a selected impedance reference point and is not interpreted as a measured physical pivot. It need not coincide with the physical contact location. The resulting contact response additionally depends on the finite rotational stiffness, the contact constraint, friction, the robot configuration, and tool-mount compliance.

2.8 Null-Space Torque

The Panda has one redundant joint direction when its 6×7 geometric Jacobian $J(q)$ has full rank. The singular value decomposition (SVD) supplies two different quantities: v_7 spans the redundant direction, while $\sigma_{\min} = \sigma_6$ indicates the weakest task-producing mapping. A thresholded pseudoinverse gives the velocity and torque projectors. The controller then combines projected joint damping with a two-probe conditioning torque.

2.8.1 SVD and Null-Space Direction

The decomposition of the current Jacobian is

$$J(q) = U_J \Sigma_J V_J^{\top}, \tag{2.55}$$

where the six task-producing singular values satisfy $\sigma_1 \geq \cdots \geq \sigma_6 \geq 0$. At full rank, the seventh right-singular vector satisfies

$$Jv_7 = 0, \tag{2.56}$$

and spans the one-dimensional kinematic null space. The conditioning indicator is

$$\sigma_{\min} = \sigma_6. \tag{2.57}$$

The vector v_7 and scalar σ_6 therefore have different roles. The first defines redundant motion; the second describes the weakest of the six task mappings. Because the geometric Jacobian mixes linear and angular rows, $\sigma_{\min}$ is used only as a relative indicator under one fixed Jacobian convention.

2.8.2 Pseudoinverse and Projectors

A relative tolerance ρ defines the retained singular values by $\sigma_i > \rho\sigma_1$. The corresponding Moore–Penrose pseudoinverse is

$$J^+ = \sum_{\sigma_i > \rho\sigma_1}^{6} \frac{1}{\sigma_i} v_i u_i^\top. \tag{2.58}$$

The joint-velocity projector extracts redundant motion,

$$N_q = I_7 - J^+ J, \qquad \dot{q}_{\text{null}} = N_q \dot{q}, \tag{2.59}$$

while the transpose projector applies a secondary joint torque without a first-order task component,

$$N_\tau = N_q^\top = I_7 - J^\top J^{+\top}. \tag{2.60}$$

With the thresholded SVD construction, the projector is orthogonal and symmetric. At full rank it reduces to $N_q = N_\tau = v_7 v_7^\top$. The implemented separation is kinematic rather than inertia weighted; Cartesian position retention is therefore checked in the pose-hold experiment.

2.8.3 Projected Damping and Conditioning

Projected null-space damping opposes the redundant joint velocity,

$$\tau_d = -d_{\text{null}} N_\tau \dot{q}. \tag{2.61}$$

The conditioning term samples two configurations along the current redundant direction. For the probe angle α_{probe}, these are

$$q_+ = q + \alpha_{\text{probe}} v_7, \qquad q_- = q - \alpha_{\text{probe}} v_7. \tag{2.62}$$

Their minimum singular values are

$$\sigma_+ = \sigma_{\min}(J(q_+)), \qquad \sigma_- = \sigma_{\min}(J(q_-)), \tag{2.63}$$

with the sampled difference

$$\Delta_\sigma = \sigma_+ - \sigma_-. \tag{2.64}$$

The direction selector applies a deadband δ_σ,

$$s_\sigma = \begin{cases} +1, & \Delta_\sigma > \delta_\sigma, \\ -1, & \Delta_\sigma < -\delta_\sigma, \\ 0, & |\Delta_\sigma| \le \delta_\sigma, \end{cases} \tag{2.65}$$

and the conditioning torque is

$$\tau_\sigma = k_\sigma N_\tau (s_\sigma v_7). \tag{2.66}$$

Sampling q_+ and q_- selects the neighbouring configuration with the larger $\sigma_{\min}$. The finite probes are local tangent approximations to the nonlinear self-motion. The commanded magnitude k_σ is independent of the probe angle.

2.8.4 Complete Null-Space Torque

The complete secondary torque is

$$\tau_{\text{null}} = \tau_d + \tau_\sigma = -d_{\text{null}} N_\tau \dot{q} + k_\sigma N_\tau (s_\sigma v_7). \tag{2.67}$$

The damping term acts only while redundant motion is present. The conditioning term is an active configuration objective and can act at zero joint velocity. Its direction is invariant to the arbitrary sign returned for v_7, because reversing v_7 also exchanges the two probes and reverses s_σ.

2.9 Complete Joint-Torque Command

The complete joint-torque command τ_{cmd} combines the Cartesian impedance torque τ_{cart}, the null-space torque τ_{null}, and the Coriolis compensation τ_c:

$$\tau_{\text{cmd}} = \tau_{\text{cart}} + \tau_{\text{null}} + \tau_c(q, \dot{q}) = J^\top(q)F + \tau_{\text{null}} + \tau_c(q, \dot{q}). \tag{2.68}$$

Here, $\tau_{\text{cart}} = J^\top(q)F$ is the Cartesian impedance torque, τ_{null} is the secondary null-space torque, and $\tau_c(q, \dot{q})$ is the Coriolis and centrifugal compensation obtained from the robot model and added in the control callback. A separate gravity term is not added because gravity compensation is handled internally by the Franka Control Interface, as described in Section 2.4.3. The next chapter implements these torque contributions in the real-time control loop and combines them with the surface-relative geometry, state-dependent references, and contact sequence.

Chapter 3

Controller Design and Real-Time Implementation

This chapter describes the real-time implementation of the Cartesian impedance controller developed in Chapter 2. The implementation combines surface-relative geometry, state-dependent reference generation, a selectable virtual CoC, null-space control, and real-time data handling. Mathematical relations already established in Chapter 2 are referenced here rather than restated, so that this chapter carries only what the controller adds to them. The controller is written in C++ using libfranka and Eigen and is executed through the Franka Control Interface (FCI) at a nominal control period of 1 ms. Numerical settings used in the experiments are given in Chapter 4 and Appendix C.

3.1 Controller Architecture and Real-Time Control Cycle

The software separates configuration and operator interaction from the real-time control calculation and from post-run data processing. The complete parameter set is read and validated before a controller session starts, so the real-time callback receives only the prepared configuration, preallocated state, the robot-model interface, and non-blocking operator requests. File output and run evaluation are performed after torque control has stopped.

At program start the FCI connection is established, error recovery is requested, the selected collision behaviour is applied, and the robot model is loaded. A separate keyboard thread receives operator requests and transfers them through atomic variables, which keeps terminal input outside the real-time callback. Reading and validating the configuration again before each session permits settings to be changed between runs without recompiling.

The operating-state flow begins with robot recovery and controller selection, as shown in Figure 3.1. A successful recovery opens the selection state. The operator can then select the surface-contact sequence, Cartesian pose hold, or manual guidance. An unsuccessful recovery ends the program. Operator stop, a configured run-duration limit, a robot-side error or reflex, and a communication or control exception provide the common transition from any active controller state to Stop.

The overview shows these three principal selections. The implementation also accepts input `t` from Controller Selection or Guidance Mode to enter the contact-impedance Test Mode directly. The experiment-oriented tuning loop is shown with the detailed contact sequence in Figure 3.3.

A robot-side error or reflex is a stop condition reported by libfranka and is not limited to a joint-limit violation. Communication and control exceptions form the separate application-side category.+

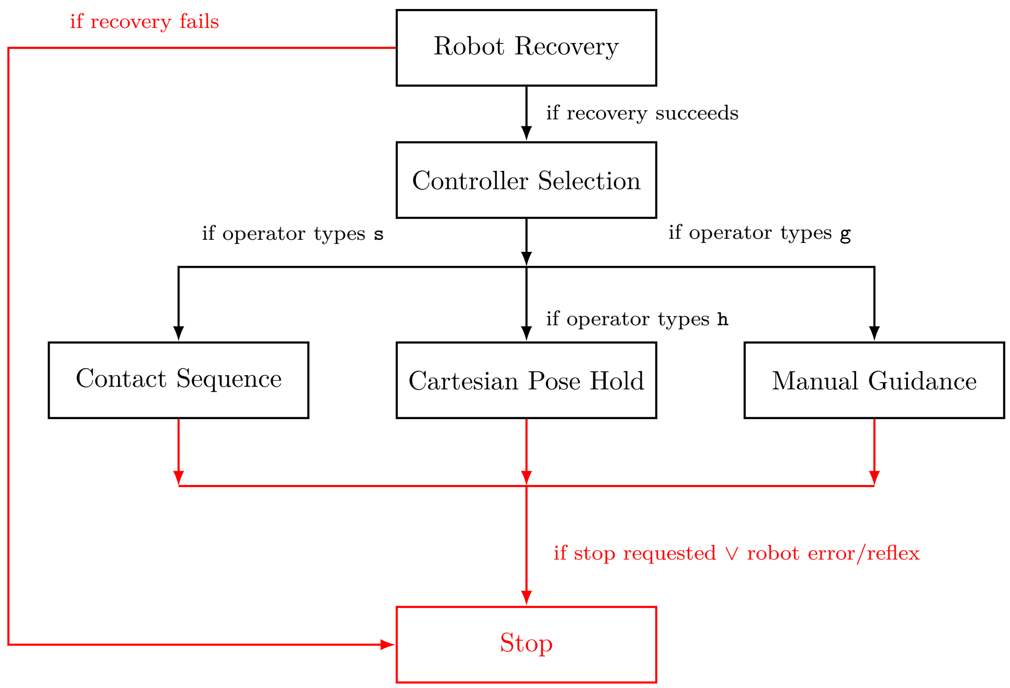


Figure 3.1: Principal controller operating-state flow.

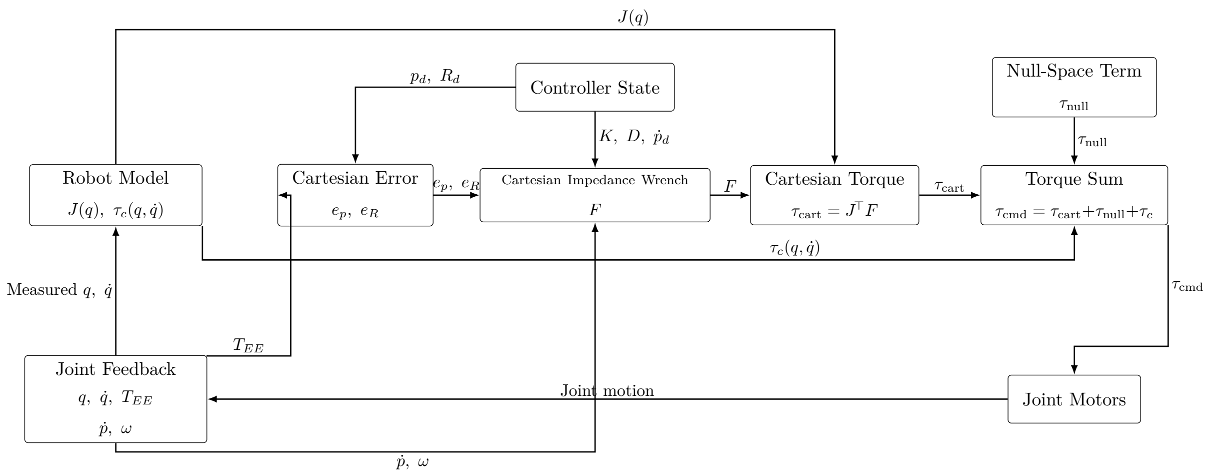


Figure 3.2: Torque contributions combined in one control cycle.

Figure 3.2 shows the torque path of one impedance-controlled cycle, and the order within that cycle is fixed. The callback first reads the measured joint configuration q, the joint velocity $\dot{q}$, the end-effector pose, the 6×7 geometric Jacobian $J(q)$, and the remaining robot-model quantities. The state logic then updates the desired Cartesian reference. The Cartesian pose error and the impedance wrench $F = [f^\top, m^\top]^\top$ are evaluated according to Section 2.2.1 and Section 2.4, using the reference of the active state and the impedance matrices selected for it. That wrench is mapped to joint torque through $J^\top(q)$, the secondary null-space torque and the model Coriolis contribution are added, the required signals are stored in preallocated memory, and the resulting torque vector is returned to libfranka.

The end-effector pose supplied by the robot state is used directly for p_{EE} and R_{EE}. The real-time controller does not reconstruct that pose from a separate Denavit–Hartenberg chain.

Two properties of the error evaluation belong to the implementation rather than to the impedance law. A rotational mask can be applied by resolving e_R in the surface frame, retaining the enabled components, and transforming the result back to the base frame; the mask acts on the rotational spring error alone, while the damping term continues to use the measured angular velocity. The controller also carries two impedance branches, one decoupled and one shifted to a virtual CoC, and the active state selects between them.

The complete impedance-controlled command follows Equation (2.68).

Here τ_{null} is the secondary null-space torque of Section 2.8.4 and $\tau_c(q, \dot{q})$ is the Coriolis and centrifugal contribution obtained from the libfranka model. No separate gravity term is added, because the FCI compensates gravity for the external joint-torque command, as described in Section 2.4.3. An experiment-specific disturbance torque is added only for the null-space pose-hold experiment of Section 4.6 and is therefore not part of the nominal controller formulation.

The implemented controller is used in two configurations in this thesis. The surface-contact configuration generates a state-dependent Cartesian reference for tool orientation, surface approach, contact establishment and the subsequent grinding state. The Cartesian pose-hold configuration instead captures and maintains a fixed Cartesian pose, and it carries the separate null-space investigation of Section 4.6. Both run the callback above, and they differ in the reference they generate and in the impedance they activate. Each is described in full before the other begins.

3.2 Surface-Contact Sequence

A surface-contact run passes through the controller states drawn in Figure 3.3. The robot may first move to the stored initial joint configuration q_{init}. Alternatively, operator input `g` enters Guidance Mode, and input `s` starts the sequence from the current guided configuration q. Input `q` stores that configuration as q_{init} for a later session. Input `p` has a different role: after Guidance Mode is entered from an active controller state, it captures the reached pose and resumes that state. Tool Orientation advances to Surface Approach after its minimum duration and either satisfaction of the configured orientation criteria or the orientation timeout. Surface Approach advances to Pre-Contact Hold when the tool clearance reaches its threshold. Reaching the maximum approach distance first instead terminates the run. The optional holds remain on the main path in the chart. A disabled hold advances without waiting, while operator confirmation releases an enabled hold. Contact Establishment advances after the moment-change criterion and minimum duration, or through its independent timeout. Pre-Grinding Hold then precedes Grinding. The reported runs ended in the enabled hold before Grinding. In the experiment-oriented loop shown, operator input `t` after Grinding opens Test Mode for the next selection of K_p, K_R, or r_c. Operator input `s` returns the sequence to initialisation. The same Test Mode can also be entered directly from Controller Selection or from Guidance Mode.

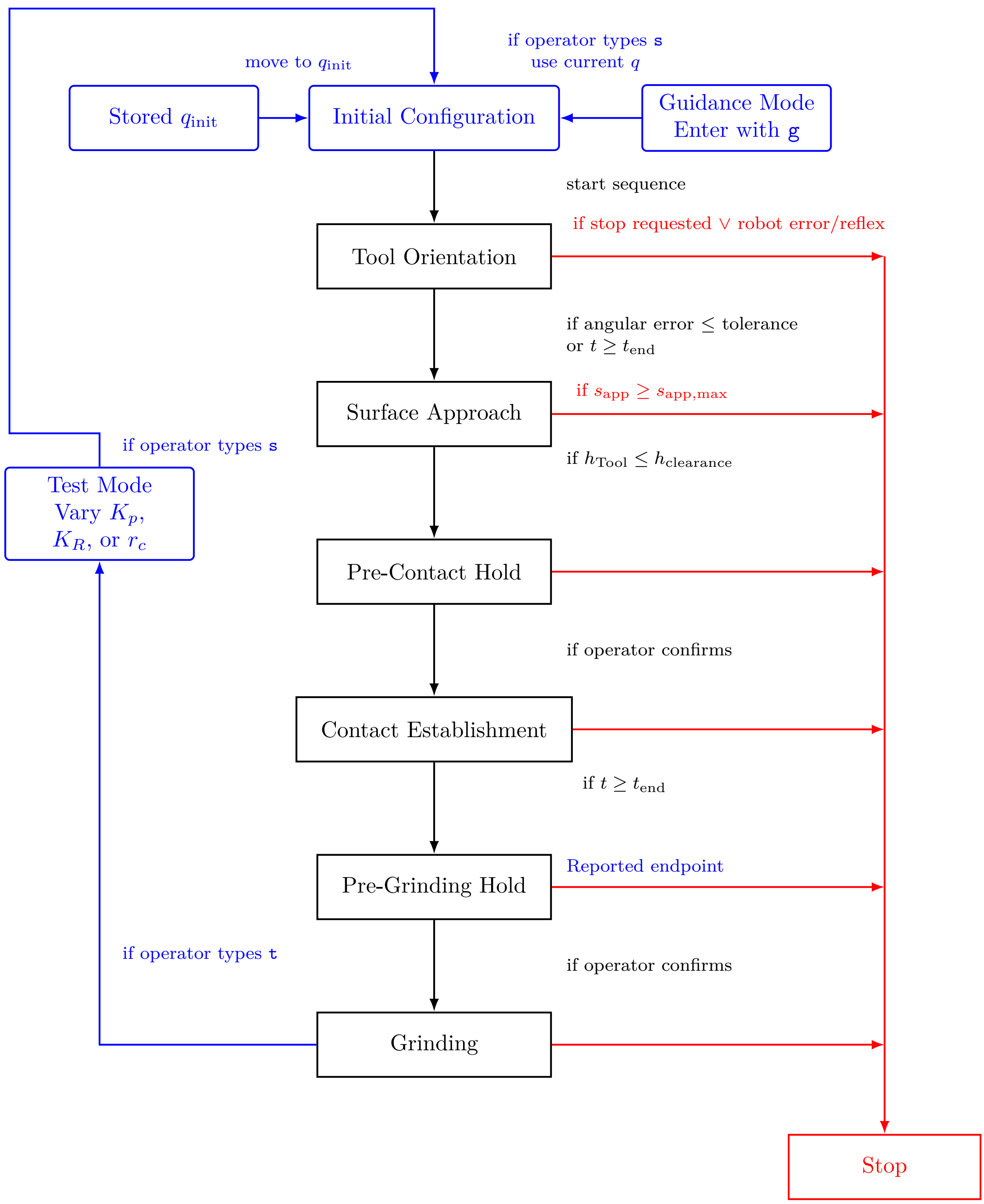


Figure 3.3: Experimental surface-contact sequence and tuning loop.

- Tool Orientation moves the tool to the commanded surface-relative angular offset while the captured TCP position is held.
- Surface Approach retains that orientation and advances the position reference towards the surface along $-n_s$. The state ends at the clearance transition, where the pose and the selected tool-point offset used to initialise contact establishment are captured.

- Operator-Controlled Hold Before Contact Establishment is optional. It retains the clearance pose until continuation is confirmed.
- Contact Establishment advances the selected tool-point reference towards the surface while the captured orientation is held. Contact-induced end-effector rotation and the optional compliance-centre coupling act in this state.
- Operator-Controlled Hold Before Grinding is optional. It retains the reached contact reference while the controller waits for confirmation.
- Grinding retains the normal press and superimposes a tangential motion. No reported run entered it.

A controller state combines a generated Cartesian reference, an active impedance configuration, and its transition conditions. Tool Orientation and Surface Approach share the first stiffness-and-damping set. Contact Establishment uses the compliant contact set and can activate the shifted virtual CoC. Grinding retains those source matrices and returns to the decoupled Cartesian impedance. The operator-controlled hold remains within the contact-establishment branch and retains its reached reference. Section 3.2.5 gives the construction, and the numerical values are stated in Chapter 4. The subsections below follow the sequence in the order the controller executes it.

3.2.1 Surface and Tool Reference

The sequence requires a frame for resolving the surface-relative directions, an orientation target for the tool face, and a geometric representation of the rectangular tool face. These quantities are defined below.

The controller constructs the surface frame according to Section 2.5 from a configured surface point p_s, the surface orientation, and a configured first-tangent hint. That hint is the in-plane reference direction of Section 2.5 and is a parameter rather than a fixed axis, so a fallback direction is applied when the configured one lies close to n_s. The resulting directions t_1, t_2 and n_s are then used consistently for the descent direction $-n_s$, the surface-relative orientation command, the directional Cartesian stiffness and damping, the compliance-centre definition when a surface-fixed displacement is selected, and the surface-resolved quantities of the experimental evaluation.

The tool normal is calibrated once in the end-effector frame. Its current direction in the base frame follows from the measured end-effector orientation:

$$n_{\text{Tool},0} = R_{\text{EE}} n_{\text{Tool,EE}}. \tag{3.1}$$

The commanded orientation offset about the two surface tangents is represented by the rotation vector

$$\theta_{\text{cmd}} = \theta_{t_1} t_1 + \theta_{t_2} t_2 \in \mathbb{R}^3. \tag{3.2}$$

Its magnitude $\|\theta_{\text{cmd}}\|$ is the commanded offset angle relative to the configured surface reference. It is a controller input and not the achieved reference-relative angle at contact entry. For a non-zero offset the corresponding unit rotation axis is

$$u_{\text{cmd}} = \frac{\theta_{\text{cmd}}}{\|\theta_{\text{cmd}}\|}, \tag{3.3}$$

and the axis–angle relation of Equation (2.7) gives the commanded rotation matrix $R_{\text{cmd}} = R(u_{\text{cmd}}, \|\theta_{\text{cmd}}\|)$, which is the identity for a zero commanded offset. The vector permits offsets about either surface tangent and along intermediate tangent-plane directions.

Since n_s points out of the surface, the tool-normal direction for a tool face parallel to the surface is $-n_s$. Applying the commanded orientation offset to that direction gives the desired tool-normal direction

$$n_d = R_{\text{cmd}} \left(-n_s\right). \tag{3.4}$$

For a zero commanded offset $n_d = -n_s$. The shortest rotation that aligns the captured tool direction with n_d is applied to the complete start orientation. The implementation can additionally prescribe a twist about the desired tool axis, which fixes the in-plane orientation of the rectangular face; because a centred rectangle is unchanged by a half-turn, the equivalent orientation closest to the captured start pose is selected. The settings used for the measurements are stated in Chapter 4.

Surface approach cannot be decided from the TCP height alone, because a face that is not parallel to the surface reaches the plane first at a corner or along an edge. The tool face is therefore represented as a rectangle fixed in the end-effector frame. Its centre offset, half-width vector, and half-length vector define the four end-effector-frame corner offsets $r_{i,\mathrm{EE}}$,

$$r_{i,\mathrm{EE}} \in \left\{ r_{\mathrm{face,EE}} \pm r_{\mathrm{width,EE}} \pm r_{\mathrm{length,EE}} \right\}. \tag{3.5}$$

The offsets are defined from the tool centre point, which coincides with the reported end-effector origin in this implementation, $p_{\mathrm{TCP}} = p_{\mathrm{EE}}$. During surface approach the four corners determine the minimum tool clearance and the geometric tool reference used for contact establishment, as described in Section 3.2.3.

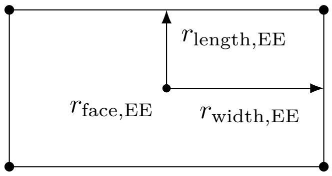


Figure 3.4: Calibrated tool face and its four corners.

3.2.2 Tool Orientation

The Tool Orientation state constructs the desired tool direction relative to the configured normal n_s. With $\theta_{\mathrm{cmd}} = 0$, the target tool face is parallel to the configured plane. A non-zero command introduces the specified angular offset about the surface tangents. The captured start position is held while the desired orientation advances towards the target of Equation (3.4) through a rate-limited interpolation. The transition uses the tool-axis error and, when enabled, the remaining spin error about the target tool axis. A minimum duration prevents immediate acceptance of a transient orientation, while a timeout supplies the alternative transition. The end-effector orientation reached at this transition is retained throughout Surface Approach.

3.2.3 Surface Approach, Clearance and Tool-Point Selection

Surface approach advances the TCP towards the plane along $-n_s$. Its scalar coordinate $s_{\mathrm{app}}(t)$ is the commanded travel from the TCP position $p_{\mathrm{TCP,start}}$ captured when the state begins at $t_{\mathrm{app},0}$,

$$s_{\text{app}}(t) = \begin{cases} v_{\text{app}}\left(t - t_{\text{app},0}\right), & t_{\text{app},0} \le t < t_{\text{app,max}}, \\ s_{\text{app,max}}, & t \ge t_{\text{app,max}}, \end{cases} \tag{3.6}$$

with the configured descent speed v_{app}, the configured maximum descent distance $s_{\text{app,max}}$, and $t_{\text{app,max}}$ the instant at which the ramp reaches that distance. Both configured values are listed in Table 4.3, and the saturation bounds an approach that does not reach the clearance threshold. The desired TCP position follows as

$$p_d(t) = p_{\text{TCP,start}} - s_{\text{app}}(t) n_s, \tag{3.7}$$

During the linear approach, the desired TCP velocity is $\dot{p}_d(t) = -v_{\text{app}} n_s$. Reaching $s_{\text{app,max}}$ before the clearance criterion is satisfied terminates the approach as unsuccessful. Where the tool face is not parallel to the surface, the corner closest to the plane can lie well below the TCP, so the TCP height alone does not give the clearance the transition needs. The controller therefore evaluates the four corners of Figure 3.4 in every control cycle. Their positions and clearances determine the geometric tool reference used to construct the contact-establishment trajectory.

The base-frame position $p_i(t)$ of tool-face corner i is

$$p_i(t) = p_{\text{TCP}}(t) + R_{\text{EE}}(t) r_{i,\text{EE}}. \tag{3.8}$$

Its signed clearance $h_i(t)$ from the configured surface reference is

$$h_i(t) = n_s^\top \left(p_i(t) - p_s\right). \tag{3.9}$$

Smaller values of h_i place a corner closer to the surface. Since p_{TCP} and p_s are common to all four corners, comparing the clearances h_i is equivalent to comparing the rotated corner offsets along the surface approach direction. The implementation uses this equivalent form for the corner comparison.

The minimum tool clearance $h_{\text{Tool}}(t)$ is the minimum signed clearance between the rectangular tool face and the configured surface reference, evaluated from its four corners:

$$h_{\mathrm{Tool}}(t) = \min_{i \in \{1,2,3,4\}} h_i(t). \tag{3.10}$$

The state ends at the clearance transition, where

$$h_{\mathrm{Tool}}(t) \leq h_{\mathrm{clearance}}, \tag{3.11}$$

with $h_{\mathrm{clearance}}$ the configured tool clearance of Table 4.3. The approach trajectory is generated for the TCP, whereas the transition to Contact Establishment is determined from the physical tool geometry. Since h_{Tool} is the minimum clearance of the four tool-face corners, $h_{\mathrm{clearance}}$ refers to the part of the rectangular tool face closest to the plane rather than to the TCP. For the reported value $h_{\mathrm{clearance}} = 0.020\,\mathrm{m}$, this part of the tool face is 20 mm above the configured reference plane at the transition. The criterion therefore provides the same specified tool clearance for different tool orientations and is purely geometric.

At the clearance transition, the same four clearances determine the selected tool point used for contact establishment. A corner belongs to the selected index set $\mathcal{I}_{\mathrm{sel}}$ when its clearance differs from the closest corner by no more than the tool-point selection tolerance $\varepsilon_{\mathrm{sel}}$:

$$\mathcal{I}_{\mathrm{sel}} = \{ i \in \{1,2,3,4\} \mid h_i - h_{\mathrm{Tool}} \leq \varepsilon_{\mathrm{sel}} \}. \tag{3.12}$$

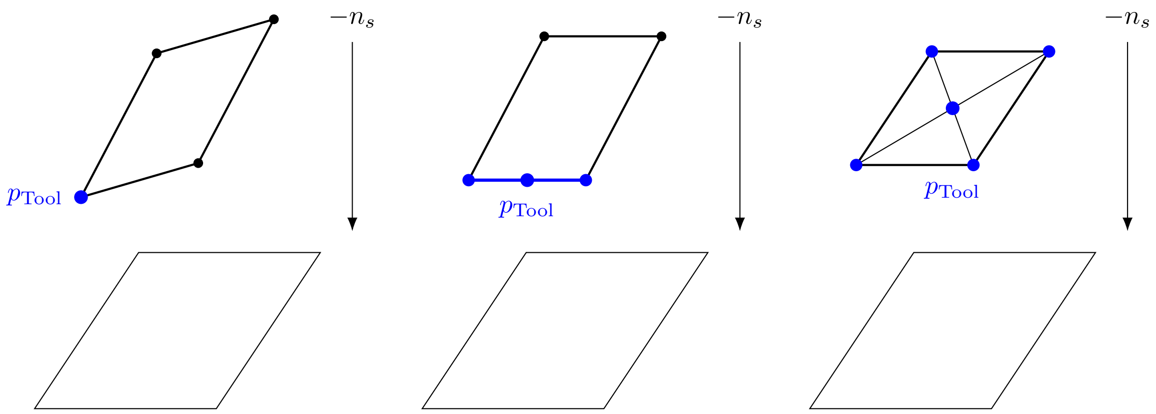
The tolerance is recorded in Table 4.1. The principal geometric cases contain one selected corner, two neighbouring selected corners, or all four corners. These give the leading corner, the leading-edge midpoint, and the tool-face centre, respectively. With $N = |\mathcal{I}_{\mathrm{sel}}|$, the selected tool-point offset $r_{\mathrm{Tool,EE}}$ is the mean of the selected corner offsets,

$$r_{\mathrm{Tool,EE}} = \frac{1}{N} \sum_{i \in \mathcal{I}_{\mathrm{sel}}} r_{i,\mathrm{EE}}. \tag{3.13}$$

The selected tool point p_{Tool} is the geometric reference used to construct the contact-establishment trajectory. It is not the instantaneous physical contact point, whose position was not measured and can change during contact. Its position is

$$p_{\text{Tool}} = p_{\text{TCP}} + R_{\text{EE}} r_{\text{Tool,EE}}. \tag{3.14}$$

The principal geometric cases are shown in Figure 3.5. For the tool-face centre, $r_{\text{Tool,EE}} = r_{\text{face,EE}}$. That calibrated offset lies along the end-effector z-axis, so the selected tool point then sits on that axis directly beneath the TCP.



(a) Leading corner (b) Leading-edge midpoint (c) Tool-face centre

Figure 3.5: Principal selected tool points for the rectangular tool face: leading corner, leading-edge midpoint, and tool-face centre.

The minimum clearance h_{Tool} determines when approach ends. The offset $r_{\text{Tool,EE}}$ determines which geometric tool point the contact-establishment reference uses. At the clearance transition the selected offset is retained in end-effector coordinates and then held constant throughout contact establishment.

Three points remain distinct in this construction. The impedance law regulates the Cartesian reference point p_{TCP}. The selected tool point p_{Tool} belongs to the physical tool geometry, and its base-frame offset $r_{\text{Tool}} = R_{\text{EE}} r_{\text{Tool,EE}} = p_{\text{Tool}} - p_{\text{TCP}}$ changes with the tool orientation. The virtual CoC p_c of Section 2.7 is a software-defined impedance reference and carries no tool geometry.

3.2.4 Operator-Controlled Hold Before Contact Establishment

The optional hold state before contact establishment maintains the captured pose until continuation is confirmed. The contact-establishment references are then captured from the current measured state.

3.2.5 Contact Establishment

In the reported surface-contact runs, contact establishment begins directly at the clearance transition. Its Cartesian reference is constructed from the selected tool point and end-effector pose captured at that instant. The captured orientation $R_{\mathrm{EE,clearance}}$ and TCP position $p_{\mathrm{TCP,clearance}}$ are taken at the clearance-transition instant $t_{\mathrm{CE},0}$,

$$R_{\mathrm{EE,clearance}} = R_{\mathrm{EE}}(t_{\mathrm{CE},0}), \qquad p_{\mathrm{TCP,clearance}} = p_{\mathrm{TCP}}(t_{\mathrm{CE},0}). \tag{3.15}$$

The corresponding selected tool-point position $p_{\mathrm{Tool,clearance}}$ is

$$p_{\mathrm{Tool,clearance}} = p_{\mathrm{TCP,clearance}} + R_{\mathrm{EE,clearance}} r_{\mathrm{Tool,EE}}. \tag{3.16}$$

Its orthogonal projection onto the configured surface reference defines the surface reference point $p_{\mathrm{Tool},0}$,

$$p_{\mathrm{Tool},0} = p_{\mathrm{Tool,clearance}} - n_s n_s^\top \left(p_{\mathrm{Tool,clearance}} - p_s\right). \tag{3.17}$$

$p_{\mathrm{Tool},0}$ is therefore the point on the configured reference plane directly below the selected tool point at the clearance transition.

Contact establishment generates the press evaluated in the experiments. The scalar coordinate $s_{\mathrm{CE}}(t)$ gives the selected tool point's commanded displacement from $p_{\mathrm{Tool},0}$ towards the surface. Its initial value is the negative signed surface coordinate of that point,

$$s_{\mathrm{CE}}(t_{\mathrm{CE},0}) = -n_s^\top \left(p_{\mathrm{Tool,clearance}} - p_s\right). \tag{3.18}$$

The selected point remains above the configured reference plane at the transition, so $s_{\mathrm{CE}}(t_{\mathrm{CE},0}) < 0$. This initial value makes the first contact-establishment reference equal to $p_{\mathrm{Tool,clearance}}$ and avoids a position step.

The coordinate advances at the configured press speed v_{CE} from $t_{\mathrm{CE},0}$ until it reaches the configured endpoint $s_{\mathrm{CE,max}}$, then remains constant:

$$s_{\mathrm{CE}}(t) = \begin{cases} s_{\mathrm{CE}}(t_{\mathrm{CE},0}) + v_{\mathrm{CE}}\left(t - t_{\mathrm{CE},0}\right), & t_{\mathrm{CE},0} \le t < t_{\mathrm{CE,max}}, \\ s_{\mathrm{CE,max}}, & t \ge t_{\mathrm{CE,max}}. \end{cases} \tag{3.19}$$

The desired selected-tool-point position $p_{\mathrm{Tool},d}(t)$ is

$$p_{\mathrm{Tool},d}(t) = p_{\mathrm{Tool},0} - s_{\mathrm{CE}}(t) n_s. \tag{3.20}$$

Appendix C records the configured press speed and endpoint.

Throughout contact establishment the desired orientation is

$$R_d(t) = R_{\mathrm{EE,clearance}}. \tag{3.21}$$

The contact-establishment trajectory is defined for the selected geometric tool point, whereas the Cartesian impedance law regulates the TCP. The desired tool-point position must therefore be converted to the corresponding TCP reference. At the retained orientation, their rigid-body relation is

$$p_{\mathrm{Tool},d}(t) = p_d(t) + R_{\mathrm{EE,clearance}} r_{\mathrm{Tool,EE}}. \tag{3.22}$$

Solving this relation gives the desired TCP position $p_d(t)$ as

$$p_d(t) = p_{\mathrm{Tool},d}(t) - R_{\mathrm{EE,clearance}} r_{\mathrm{Tool,EE}}. \tag{3.23}$$

During the linear part of the ramp, the desired TCP velocity is $\dot{p}_d = -v_{\mathrm{CE}} n_s$. It is zero after $s_{\mathrm{CE,max}}$ is reached. The controller generates the press by prescribing the selected tool-point trajectory and reconstructing the TCP target that realises it. Because the surface restricts this commanded motion, the resulting Cartesian position error generates the press. The configured endpoint lies below the configured reference plane and defines a spring reference rather than an expected physical travel.

Figure 3.6 shows the complete construction from the projected tool point to the TCP target.

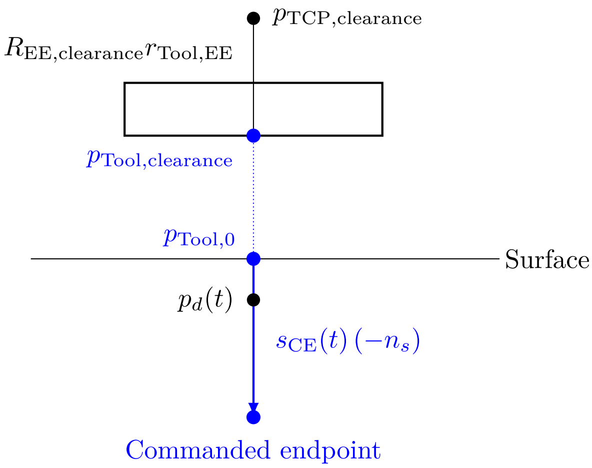


Figure 3.6: Contact-establishment reference generation.

No time-varying tipping trajectory is commanded during contact establishment. The held orientation reference acts as a finite-stiffness rotational spring rather than as a rigid pose constraint, so finite rotational compliance permits contact-induced end-effector rotation. When the virtual CoC is enabled, the point-shifted matrices of Section 2.7 additionally couple the translational press to the commanded moment; the same contact-establishment position reference is used in both cases.

The model-estimated external wrench supplied by libfranka is stored at the clearance transition for the optional termination condition. After the configured minimum duration, contact establishment can terminate when the estimated external moment change exceeds the configured threshold, or through the independent timeout. Every reported contact run terminated through the timeout, so that threshold did not affect the reported results. During the operator-controlled hold before grinding, the controller remains in the contact-establishment branch while waiting for confirmation. The coupled impedance stays active where it was selected.

The impedance acting on that reference changes at the same transition, from the approach set to the contact-establishment set. Three choices determine it: the directional stiffness, the damping associated with those directions, and whether the impedance reference point is the TCP or a shifted virtual CoC. The controller constructs the directional stiffness and damping first and applies the compliance-centre transformation afterwards.

Directional Cartesian stiffness and damping are defined relative to the surface frame and transformed to the robot base frame by the congruence of Section 2.6. Each state of the sequence activates its own stiffness and damping set in the configured surface frame, and the contact-establishment translational frame remains configurable. Contact Establishment and Grinding draw on the same source stiffness and damping blocks; the state transition changes the generated Cartesian reference and the choice between the coupled and decoupled branch. An operator-controlled hold can temporarily apply dedicated hold stiffness and damping while maintaining the current reference. The numerical values used in the experiments are given in Chapter 4.

For the states using inertia-scaled damping, the directional coefficients are calculated from Equation (2.33). The implementation obtains the joint-space inertia $M(q)$ and Jacobian $J(q)$ from the libfranka model. An LDLT factorisation solves the required system without explicitly forming $M(q)^{-1}$. The result is cached for each stiffness-and-damping configuration and recalculated when that configuration is activated. Coefficient bounds, validity checks, and the fallback damping matrix are recorded in Appendix C.

The reported surface-contact experiments used inertia-scaled damping during Surface Approach and Contact Establishment.

The point-shift formulation derived in Section 2.7 is implemented with the displacement $r_{c,0} = p_c - p_{\mathrm{TCP}}$. Two definitions are supported, because the meaning of a fixed displacement depends on the frame that holds it.

A tool-fixed displacement is configured in end-effector coordinates. Its base-frame vector $r_{c,0} = R_{\mathrm{EE}} r_{c,\mathrm{EE}}$ is therefore updated from the measured end-effector orientation in every callback and rotates as the end effector rotates. A surface-fixed displacement is configured in surface coordinates, so $r_{c,0} = R_{\mathrm{surface}} r_{c,S}$ stays constant during contact establishment, the surface frame being fixed for one controller session. The configuration loader requires exactly one of the two definitions to be active when the virtual centre is enabled, and rejects a parameter set selecting both or neither. Only the tool-fixed definition in end-effector coordinates was used in the reported experiments.

The source translational and rotational matrices are assembled into block-diagonal spatial stiffness and damping at p_c. The stiffness and damping are shifted according to Equations (2.38) and (2.39).

For a tool-fixed displacement the shifted base-frame matrices are updated whenever the measured end-effector orientation changes, whereas a surface-fixed displacement leaves them constant as long as the source matrices do not change. Setting $r_{c,0} = 0$ makes the adjoint the identity and returns the decoupled block-diagonal form.

During contact establishment this point-shifted impedance allows the commanded normal press to contribute to the rotational response. The influence of the compliance-centre position on that response is evaluated experimentally in Chapter 5. The coupled branch is used only in the controller states configured for a virtual CoC; the cases that used it are specified in Chapter 4 and Appendix C.

3.2.6 Operator-Controlled Hold Before Grinding

The hold state begins when Contact Establishment satisfies its moment-change condition after the minimum duration, or when its timeout is reached. The controller retains the reached contact-establishment reference and waits for operator confirmation. Every reported surface-contact run ended in this state, and Grinding was not entered.

3.2.7 Grinding

After operator confirmation, the Grinding state retains the normal press and superimposes a smooth tangential motion using the decoupled impedance. No reported run entered this state.

3.3 Cartesian Pose Hold

Cartesian pose hold starts after the robot moves to the stored initial joint configuration q_{init}, or directly from the current guided configuration q when the operator types `h` in Guidance Mode. The current end-effector pose is then captured at the entry instant t_0 and used to define the fixed references $p_d = p_{\mathrm{EE}}(t_0)$ and $R_d = R_{\mathrm{EE}}(t_0)$. Its dedicated hold impedance is expressed in the base frame. Surface Approach, clearance detection and contact-establishment reference generation take no part in it. The reported trials used fixed hold damping rather than the inertia-scaled rule of Section 2.6.1.

The operator selects the secondary-torque mode by entering 0, 1, 2, or 3. Mode 0 commands no null-space torque. Mode 1 applies projected damping with adjustable d_{null}. Mode 2 applies singular-value conditioning with adjustable k_σ. Mode 3 applies both terms and permits both parameters to be varied. Entering another number switches directly to the selected mode while Cartesian pose hold remains active. The Cartesian reference and Cartesian impedance remain unchanged.

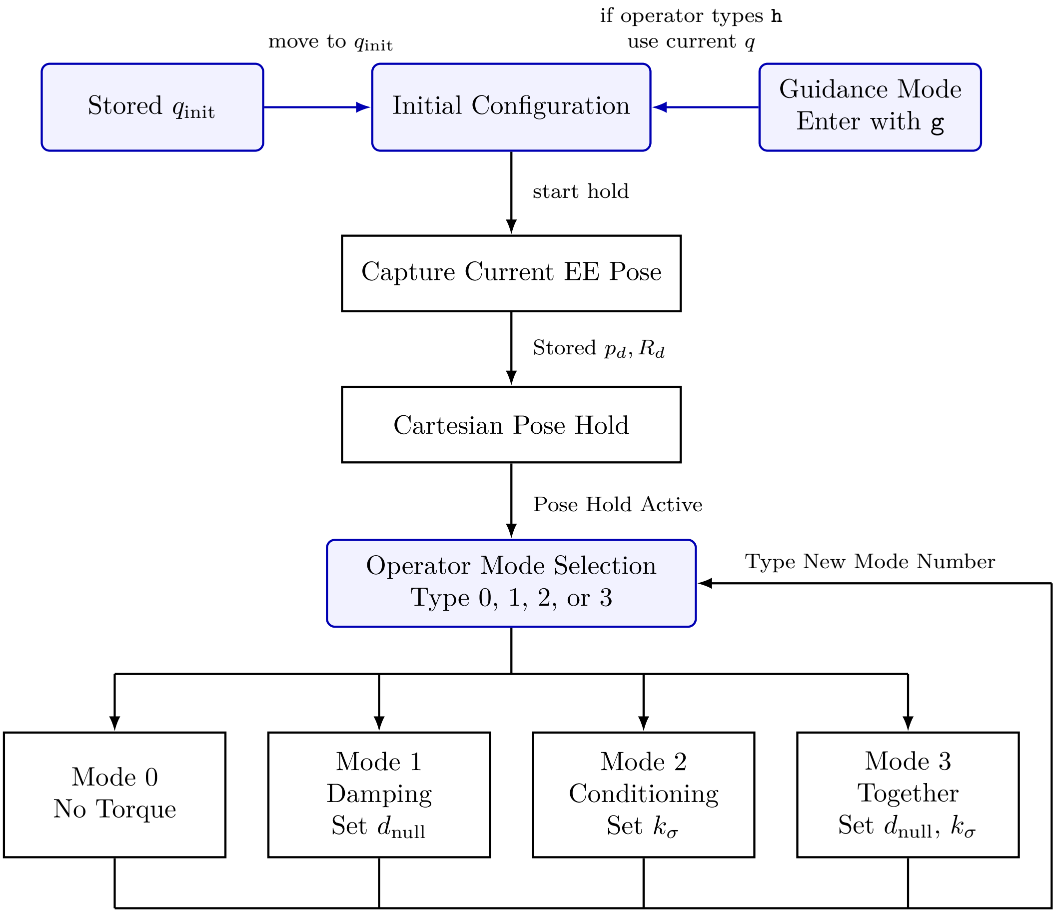


Figure 3.7: Operator-selectable null-space modes in Cartesian pose hold.

The four reported pose-hold settings were mode 0, mode 1, and mode 2 at two values of k_σ. The surface-contact sequence used mode 3, which was not compared in pose hold. The disturbance, its timing and the evaluated quantities belong to Section 4.6.

The implementation adds two choices not fixed by Section 2.8. The pseudoinverse threshold is the configured relative tolerance multiplied by the largest current singular value. The resulting joint-torque projector is symmetrised numerically, and the same matrix produces the recorded projected joint velocity $\dot{q}_{\text{null}}$. For singular-value conditioning, the controller evaluates the Jacobian at $q \pm \alpha_{\text{probe}} v_7$ and compares the smallest singular value at each probe. This comparison uses the current robot transformations, and no conditioning torque is requested inside the configured deadband. The surface-contact sequence holds one combined setting throughout, so the secondary controller is not a variable in those comparisons. Every numerical value is given in Chapter 4 and Appendix C.

3.4 Robot-Side Safety and Command Handling

The joint-torque command of Equation (2.68) is returned directly to libfranka. No application-side saturation or torque-rate limiter is applied after the Cartesian, null-space and Coriolis terms have been assembled, so the configured collision and reflex monitoring is the independent protection layer during the reported experiments. Its joint-torque and Cartesian-wrench thresholds are recorded in Section 4.1.3. A robot communication, motion-generation or torque-control error leaves the callback through `franka::Exception` and ends the control session.

Everything the robot needs before torque control is completed before it starts: the connection, error recovery, the collision configuration, the configuration validation, the gripper action and the logging-buffer allocation. A libfranka joint-motion generator then brings the robot to a configured initial joint posture, or the run starts from a pose captured after manual guidance, with the posture used in the reported experiments recorded in Appendix C. Operator input and file writing are handled outside the real-time thread, so neither adds a blocking operation to the joint-torque callback.

3.5 Real-Time Data Recording

Experimental signals are stored during torque control in a preallocated in-memory ring buffer, so the callback performs no file access and grows no buffer dynamically. Each sampled row carries the measured state, the active reference, the calculated error and wrench, the controller state, the secondary-controller configuration, and the returned joint torque. When the allocated capacity is reached the write index wraps and new rows replace the oldest, which keeps the memory bounded while control continues. After the callback ends, the retained samples are written to the CSV file in chronological order.

Values captured at the clearance transition refer to the geometric event of Section 3.2.3. The complete column-level description is given in Appendix B, and the experimental metrics calculated from these signals are defined in Chapter 4.

Chapter 4 next specifies the physical system, the calibration procedure, the experimental parameter matrix, the contact procedure, and the evaluation quantities used to assess this controller.

Chapter 4

Experimental Methodology

This chapter describes how the controller developed in Chapter 3 was evaluated experimentally. The methodology is organised in the order in which the experiments can be reproduced: the physical system is introduced first, followed by the surface reference and tool calibration, the common contact-test configuration and procedure, the main surface-contact matrix, the quantities used for evaluation, and the separate Cartesian pose-hold study for the null-space controller. Supporting checks of the compliance-centre geometry are reported in Appendix D.

The main data set consists of surface-contact measurements in which the commanded tool orientation offset, directional Cartesian stiffness, and virtual CoC were varied under otherwise matched conditions. The primary response quantity is the signed contact-establishment response. A second, independent experiment evaluates the null-space terms under a repeatable internally commanded disturbance.

4.1 Experimental System

4.1.1 Robot, Tool, and Surface

The experiments were performed with a seven-degree-of-freedom Franka Emika Panda operated through the Franka Control Interface. Joint-torque commands were transmitted through libfranka at the nominal control rate of 1 kHz, corresponding to a sampling period of 1 ms. The real-time controller, Cartesian impedance law, contact geometry, and state sequence are described in Chapter 3.

The rectangular grinding tool was held by the Franka Hand using custom gripper fingers. The tool face measured 40×120 mm, with the width along X_{EE} and the length along Y_{EE}. Its centre was located 20 mm along $+Z_{\mathrm{EE}}$ from the reported end-effector origin. The tool was clamped by custom pads screwed to the gripper fingers. Mechanical clearance in that clamping allowed the tool to rotate relative to the gripper by approximately $\pm 2°$ about y_{EE}, which corresponds approximately to the t_2 direction in the flat experimental configuration. The robot does not measure this relative rotation: it is absent from the joint and end-effector state. The $\pm 2°$ was measured in the unloaded condition, and the instantaneous relative tool–gripper rotation was not tracked during the contact experiments. It is therefore a mechanical property of the mounting rather than a calibrated measurement uncertainty.

Orientation quantities reconstructed from R_{EE} and the calibrated tool normal consequently describe the tool orientation under the assumption of a fixed tool-to-end-effector transformation, and any motion of the tool within the gripper is not included in them.

The surface was a plane plate mounted on a raised base board. All reported measurements used dry contact. The configured surface reference and the tool normal were determined separately before the measurements, as described in Section 4.2.

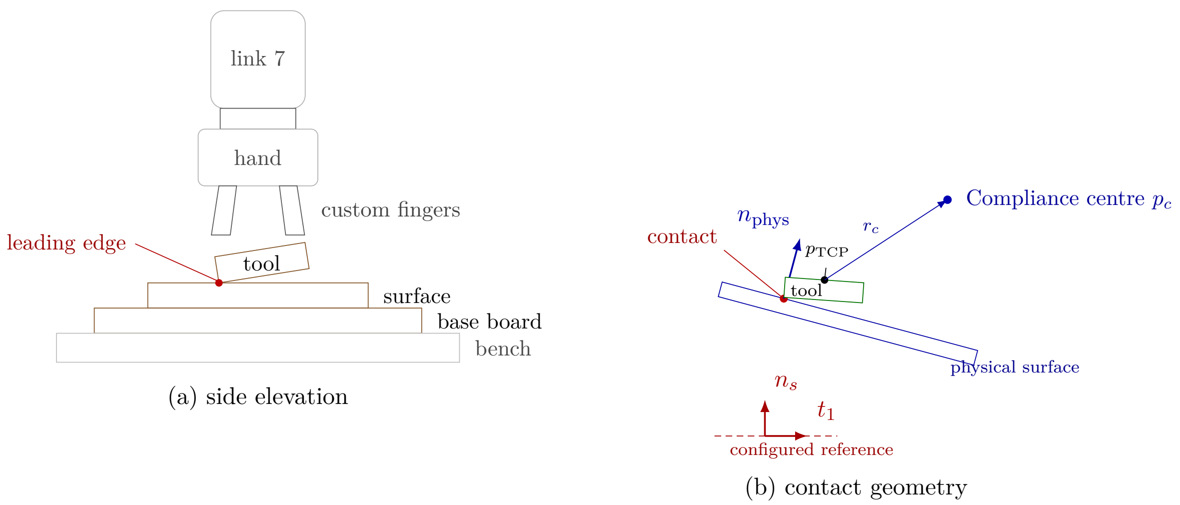


Figure 4.1: Experimental system and surface-frame contact geometry.

4.1.2 Real-Time Computer and Software

The controller computer ran Ubuntu 20.04.3 with the `5.9.1-rt20` Linux kernel and the `PREEMPT_RT` patch. The controller was compiled using `g++ 9.3`, C++14, and optimisation level `-O2`. Eigen 3.3.7 was used for matrix calculations and libfranka 0.7.1 for communication with the robot.

Before the measurements, the computer–robot connection was checked with the `communication_test` program supplied with libfranka 0.7.1 [6]. The program reports the fraction of control commands received by the robot and issues its lower communication-quality warning below an average command-success rate of 0.90. The experimental system passed this criterion. This check verifies communication quality for operation through the nominal 1 kHz interface; it is not a worst-case execution-time or scheduling-jitter measurement.

4.1.3 Operating and Safety Configuration

Before torque control started, the robot-side collision behaviour was configured with equal acceleration and nominal joint-torque thresholds of 140 N m. The six Cartesian force and moment channels used thresholds of 240 N or 240 N m, according to the component type. These thresholds supplied the Franka reflex mechanism and were kept unchanged throughout the reported measurements.

The complete run-specific controller configuration is given in Appendix C.

4.2 Configured Surface Reference, Physical Surface, and Tool Calibration

Four quantities distinguish the controller reference from the physical contact geometry. The configured normal n_s is available to the controller, whereas n_{phys} denotes the unmeasured physical surface normal. The command θ_{cmd} sets the desired tool offset relative to n_s. The pose-based angle θ_0 is the reference-relative orientation actually reached at contact entry.[+]

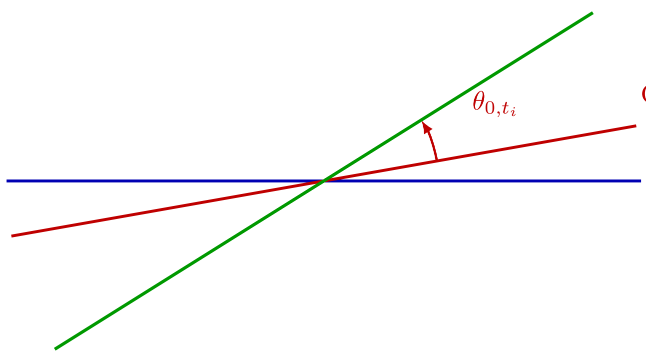


Figure 4.2: Physical, configured, and contact-entry angular references.

Two independent procedures were performed before the contact experiments. The surface-reference construction determined p_s, n_s, and R_{surface} in the robot base frame. The tool-normal calibration determined $n_{\text{Tool,EE}}$ in end-effector coordinates. The procedures are described separately below.[+]

The tool-normal calibration was not used to revise the configured surface reference. Consequently, n_s represents the geometry available to the controller rather than an independent measurement of the physical surface normal. The physical surface-reference difference was not measured independently.[+]

4.2.1 Surface-Reference Construction

The complete tool face was seated on the plane and one end-effector pose was recorded. The nominal $+Z_{\text{EE}}$ direction defined the configured surface normal n_s; the tool-face-centre offset defined p_s. The tangents were constructed as in Section 2.5. Stored surface tilts were $-1.585°$ about base x and $+0.988°$ about base y. See Appendix C.[+]

4.2.2 Tool Normal Calibration

The complete tool face was seated at three yaw orientations. The direction in end-effector coordinates that remained most nearly invariant in the base frame was fitted from these captures, yielding $n_{\mathrm{Tool,EE}}$. The calibrated normal differed from nominal $+Z_{\mathrm{EE}}$ by approximately 1.56°. The fitting relation and numerical vector are given in Appendix C.+

4.2.3 Calibrated Experimental Geometry

The rectangular face geometry used by the controller is listed in Table 4.1. The selected-tool-point construction and the reconstruction of the TCP target are described in Section 3.2.3 and Section 3.2.5, respectively.

Table 4.1: Calibrated tool geometry.

Quantity	Value in end-effector coordinates
Tool-face centre	$[0, 0, 0.020]^{\top}$ m
Half-width	$[0.020, 0, 0]^{\top}$ m
Half-length	$[0, 0.060, 0]^{\top}$ m
Tool-point selection tolerance, $\varepsilon_{\mathrm{sel}}$	0.1 mm

4.3 Common Contact-Test Configuration and Procedure

All surface-contact cases used the same calibrated geometry, state sequence, normal press trajectory, fixed stiffness and damping entries, and null-space configuration except where a parameter was explicitly varied. This section defines that common experimental basis before the individual cases are introduced in Section 4.4.

4.3.1 Common Cartesian Stiffness and Damping

The state-dependent stiffness and damping matrices are diagonal in the configured surface frame $[t_1, t_2, n_s]$ before any compliance-centre shift. Table 4.2 gives the entries used by every reported contact run, together with the damping ratio ζ of the corresponding state. The optional operator-controlled hold before Contact Establishment used the same stiffness as Contact Establishment. It was disabled in the reported runs and therefore was not entered.

Table 4.2: Common Cartesian stiffness and damping ratio by state.

	K_p [N/m]			K_R [N m/rad]			
State	t_1	t_2	n_s	t_1	t_2	n_s	ζ
Orientation and approach	150	150	150	180	180	90	1.9
Contact-establishment	2000	2000	350	5	5	50	1.0

During Contact Establishment, the lower rotational stiffness about t_1 and t_2 permits contact-induced end-effector rotation, while the higher stiffness about n_s resists in-plane spin. The directional damping coefficients were calculated using the inertia-scaled formulation of Section 2.6.1.

4.3.2 Contact Sequence and Contact-Establishment Motion

The common trajectory and transition parameters are collected in Table 4.3. These settings were used throughout the surface-contact experiments unless explicitly stated otherwise.

Table 4.3: Common state-transition and trajectory parameters.

Quantity	**Value**	**Role**
Minimum orientation time	1.0 s	Prevents an immediate state transition.
Tool-axis transition tolerance	0.016 rad	Maximum remaining tool-axis error before descent.
Spin transition tolerance	0.009 rad	Maximum remaining normal-axis spin error.
Orientation timeout / maximum rate	8.0 s / 20°/s	Bounds Tool Orientation.
Descent speed	0.1 m/s	Reference speed along the descent direction.

Table 4.3: Common state-transition and trajectory parameters (continued).

Quantity	Value	Role
Maximum descent distance	0.640 m	Terminates an unsuccessful approach.
Tool clearance, $h_{\text{clearance}}$	0.020 m	Minimum tool-face clearance at the transition to Contact Establishment.
Minimum contact-establishment time	2.0 s	Earliest activation of the moment-based exit condition.
Contact-establishment timeout	5.0 s	Time-based contact establishment termination.
Moment-change threshold	150 N m	Alternative contact establishment termination based on the model-estimated external moment.
Contact-establishment push speed	0.080 m/s	Rate of the signed press coordinate.
Contact-establishment push endpoint	+0.240 m	Final signed Cartesian spring reference.
Operator-controlled hold before Contact Establishment	disabled	Surface Approach transfers directly to Contact Establishment.
Operator-controlled hold before Grinding	enabled	Holds the reached pose after Contact Establishment.

Before each run, the active parameter set was loaded and the robot was moved to the configured initial posture. Surface Approach ended when the minimum rectangular-tool clearance reached $h_{\text{clearance}}$. Contact Establishment then advanced the Cartesian spring reference while retaining the orientation captured at the clearance transition. Every reported run ended at the 5 s timeout and subsequently entered the operator-controlled hold before Grinding.+

Projected null-space damping and singular-value conditioning remained active together throughout these measurements, with $d_{\text{null}} = 2\,\text{N}\,\text{m}\,\text{s/rad}$ and $k_\sigma = 1\,\text{N}\,\text{m}$.+

4.3.3 Repetitions and Data Recording

The recorded signals and CSV structure are documented in Section 3.5 and Appendix B.

Three repetitions were recorded for each parameter setting. Results are reported as arithmetic means, with sample standard deviations given for the main comparisons and where relevant in the supporting checks. The main surface-contact study comprises 111 runs across 37 settings. A further 45 runs across 15 settings were performed as supporting checks and are reported in Appendix D. In total, 156 surface-contact runs across 52 settings were recorded. No run was discarded. Together with the 12 pose-hold runs of Section 4.6, the complete experimental data set contains 168 recorded runs.

Within each controlled comparison, the calibrated geometry, common state parameters, fixed stiffness and damping, and null-space settings remained unchanged. Only the parameter assigned to the corresponding comparison was varied.

4.4 Main Surface-Contact Experimental Matrix

The main surface-contact study separates the effects of the principal controller parameters. Case A establishes the TCP-centred reference, where $r_c = 0$. In Cases B and C, rotational and translational stiffness were varied independently. This allowed the effect of either entry to be evaluated while the other remained fixed. In Case D the tangential compliance-centre position was varied across both principal surface tangents and both signs of the pose-based initial angular deviation.

All non-zero compliance-centre displacements were configured and held constant in end-effector coordinates. The tangential coordinates reported for Case D are the surface-frame projections of those settings in the flat target orientation.

The initial angular-deviation magnitude comparison, displacement along the tool axis, and intermediate tangent-plane directions are treated as supporting checks in Appendix D.

Table 4.4 summarises the parameter variations of the main study. Each listed setting was repeated three times unless it is a reference condition counted with an earlier case.

Table 4.4: Main surface-contact cases and parameter variations.

Case	Varied quantity	Pose-based initial angular deviation	Tested values	Runs	Evaluated effect
A	CoC at the TCP	0.74°; +9.31° and −9.41° about t_1; +10.05° and −9.55° about t_2	$r_c = 0$.	15	Contact response without CoC coupling.
B	Rotational stiffness	+9.31° to +9.32° about t_1; +10.02° to +10.05° about t_2	Command-axis $K_R = 15$ and $50\,\mathrm{N\,m/rad}$.	12	Influence of rotational stiffness.
C	Translational stiffness	+9.24° to +9.31° about t_1; +10.02° to +10.05° about t_2	Cross-axis $K_p = 300$ and $800\,\mathrm{N/m}$.	12	Influence of translational stiffness.
D	Tangential CoC position	About +9.32° and −9.36° about t_1; about +10.03° and −9.56° about t_2	−40, −20, −10, +10, +20, +40 mm along the tangent perpendicular to the commanded rotation.	72	Direction and sign dependence.

The run counts in Table 4.4 give the three scheduled repetitions of each setting. A condition that a case shares with an earlier one is counted with that earlier case.

In Cases B and C, only one Cartesian stiffness entry was varied within each comparison. The remaining stiffness and damping entries were kept at the common values of Table 4.2.

4.5 Experimental Condition and Response Quantity

The commanded tool-orientation offset θ_{cmd} is the controller input used to produce the pose-based initial angular deviation θ_{0,t_i}. Contact Establishment then produces the signed response γ_{t_i}, which is used for the main experimental comparisons.

4.5.1 Pose-Based Initial Angular Deviation

At the start of Contact Establishment, the calibrated tool normal in base coordinates is $n_{\mathrm{Tool},0}(t_{\mathrm{CE,start}})$. Let the rotation vector $e_0 = \phi_0 u_0$ denote the shortest rotation from this direction back to the configured flat direction $-n_s$, so that

$$R(u_0, \phi_0) n_{\text{Tool},0}(t_{\text{CE,start}}) = -n_s. \tag{4.1}$$

The pose-based initial angular-deviation vector is defined in surface coordinates as

$$\theta_{0,S} = -R_{\text{surface}}^{\top} e_0 = \begin{bmatrix} \theta_{0,t_1} & \theta_{0,t_2} & \theta_{0,n} \end{bmatrix}^{\top}. \tag{4.2}$$

The minus sign makes a component directed away from the configured flat orientation carry the same sign as the command that produced it. This angle is calculated from the measured end-effector pose and the calibrated tool normal. It is therefore the achieved reference-relative condition rather than an independent measurement of the physical tool–surface angle.

4.5.2 Signed Contact-Establishment Response

Let $t_{\text{CE,start}}$ and $t_{\text{CE,end}}$ denote the measurement instants at the beginning and end of contact establishment. The corresponding measured end-effector orientations relative to the robot base frame are

$$R_{\text{CE,start}} = R_{\text{EE}}(t_{\text{CE,start}}), \qquad R_{\text{CE,end}} = R_{\text{EE}}(t_{\text{CE,end}}). \tag{4.3}$$

Contact-establishment holds the orientation captured at the clearance transition as its rotational reference, and the measured orientation coincides with it at the start of the state, so $R_{\text{CE,start}}$ is that held reference. The relative rotation from the measured end orientation back to it is therefore

$$R_{\text{CE}} = R_{\text{CE,start}} R_{\text{CE,end}}^{\top}. \tag{4.4}$$

Writing this relative rotation in axis–angle form as $R_{\text{CE}} = R(u_{\text{CE}}, \phi_{\text{CE}})$ gives the measured contact-establishment rotation vector γ_0 in base coordinates:

$$\gamma_0 = \phi_{\text{CE}} u_{\text{CE}} = e_R \Big|_{\text{end of Contact Establishment}}. \tag{4.5}$$

The calculation uses only the measured end-effector orientations and therefore does not require the instantaneous tool-to-gripper rotation, which was not tracked during the contact experiments. The surface frame enters solely as the directions the rotation is resolved along:

$$\gamma_S = R_{\text{surface}}^{\top}\gamma_0 = \begin{bmatrix}\gamma_{t_1} & \gamma_{t_2} & \gamma_n\end{bmatrix}^{\top}, \qquad \gamma_{t_i} = t_i^{\top}\gamma_0, \quad i \in \{1, 2\}. \tag{4.6}$$

The reported quantity γ_{t_i} is the signed contact-establishment response about surface tangent t_i. It represents the end-of-contact-establishment orientation change relative to the held contact-establishment reference, resolved along that tangent. Its sign is chosen such that a response reducing the initial angular deviation θ_{0,t_i} carries the same sign as that deviation. A correcting response is positive for a positive initial deviation and negative for a negative one, and its magnitude is the size of the correction. Reading the sign alone is therefore not enough, since θ_{0,t_i} fixes which direction counts as correcting. The response is primary because it is obtained directly from the measured end-effector orientations and requires no independent measurement of the physical surface normal or tool orientation. The pose-based and geometry-based consistency checks are reported in Appendix D.6 and are not used for the main comparisons.

4.6 Secondary Null-Space Pose-Hold Experiment

The surface-contact measurements keep projected damping and singular-value conditioning active together. A separate Cartesian pose-hold experiment was therefore used to compare their individual effects under a repeatable internally commanded disturbance.

4.6.1 Experimental Configuration

The pose-hold controller maintained the initial TCP pose with the stiffness and damping of Table 4.5. Every entry was isotropic, so each matrix is the stated value times the identity. Unlike the contact-sequence states, the twelve pose-hold trials used fixed Cartesian damping rather than inertia-scaled damping.

Table 4.5: Isotropic Cartesian stiffness and damping of the pose-hold trials.

Quantity		**Value**
Translational stiffness	$K_{p,\mathrm{hold}}$	$2000\,\mathrm{N/m}$
Rotational stiffness	$K_{R,\mathrm{hold}}$	$50\,\mathrm{N\,m/rad}$
Translational damping	$D_{p,\mathrm{hold}}$	$50\,\mathrm{N\,s/m}$
Rotational damping	$D_{R,\mathrm{hold}}$	$8\,\mathrm{N\,m\,s/rad}$

Four null-space settings were tested with three repetitions each: no null-space torque, projected damping with $d_{\mathrm{null}} = 2\,\mathrm{N\,m\,s/rad}$, sigma conditioning with $k_\sigma = 1.5\,\mathrm{N\,m}$, and sigma conditioning with $k_\sigma = 2.0\,\mathrm{N\,m}$. The probe step, sigma deadband, and relative SVD threshold remained at $0.08\,\mathrm{rad}$, 10^{-6}, and 10^{-4}, respectively.

4.6.2 Commanded Disturbance

The disturbance represents the joint-torque equivalent of a commanded point force acting at a specified location on link 3. The point was defined by the link-frame offset

$$r_p = \begin{bmatrix} 0 \\ 0 \\ 0.200 \end{bmatrix} \mathrm{m}. \tag{4.7}$$

At the initial posture q_{init}, the structural null direction $v_7(q_{\mathrm{init}})$ and the translational point Jacobian $J_p(q_{\mathrm{init}})$ define the fixed base-frame force direction

$$u_f = \frac{J_p(q_{\mathrm{init}})v_7(q_{\mathrm{init}})}{\|J_p(q_{\mathrm{init}})v_7(q_{\mathrm{init}})\|_2}. \tag{4.8}$$

The commanded force magnitude was $F_d = 20\,\mathrm{N}$. The virtual force and its equivalent joint torque were

$$f_d(t) = F_d s_d(t) u_f, \qquad \tau_{\mathrm{dist}}(t) = J_p(q(t))^\top f_d(t). \tag{4.9}$$

Here u_f is the fixed unit force direction defined in Equation (4.8), and $s_d(t) \in [0, 1]$ is a dimensionless time-scaling function that switches the commanded virtual disturbance smoothly on and off. It was defined as

$$
s_d(t) = \begin{cases} 0, & t < 5\,\mathrm{s}, \\ \dfrac{1}{2}\left[1 - \cos\!\left(\pi \dfrac{t - 5\,\mathrm{s}}{2\,\mathrm{s}}\right)\right], & 5\,\mathrm{s} \le t < 7\,\mathrm{s}, \\ 1, & 7\,\mathrm{s} \le t < 8\,\mathrm{s}, \\ \dfrac{1}{2}\left[1 + \cos\!\left(\pi \dfrac{t - 8\,\mathrm{s}}{1\,\mathrm{s}}\right)\right], & 8\,\mathrm{s} \le t < 9\,\mathrm{s}, \\ 0, & t \ge 9\,\mathrm{s}. \end{cases} \tag{4.10}
$$

Thus the disturbance increased smoothly from zero to the full commanded magnitude between 5 and 7 s, remained at full magnitude until 8 s, and decreased smoothly to zero by 9 s. The equivalent disturbance torque was limited to

$$
\|\tau_{\mathrm{dist}}\|_2 \le 3\,\mathrm{N\,m}.
$$

The disturbance is internal to the controller and therefore provides a repeatable comparison input; it is not a separately applied physical force.

The equivalent torque is formed from the point Jacobian alone and carries no null-space projector, so it is a joint-torque excitation rather than a null-space command. Its image therefore has a component along the six Cartesian directions as well as along the redundant direction, and the Cartesian impedance resists that component while the disturbance acts. Equation (4.11) therefore checks that the commanded pose was retained rather than assuming it.

In every trial the selected null-space law was active from the start of the pose-hold state. The first 5 s therefore formed a pre-disturbance settling interval, during which the conditioning torque of the sigma-only settings could already move the redundant configuration. The disturbance then acted from 5 to 9 s, followed by an observation interval until the end of the trial. No secondary term was enabled or disabled within a trial: the sigma-conditioning torque remained active throughout the complete sigma-only trial and was not switched on after the disturbance had been removed.

4.6.3 Evaluation Quantities

All quantities in this subsection are evaluated over the interval in which the internally commanded virtual disturbance acts, $t \in [5, 9]\,\mathrm{s}$. The first criterion checks whether the primary Cartesian pose-hold objective is maintained while the disturbance and the selected null-space law act. The predefined position-retention criterion is

$$\max_{5\,\mathrm{s} \le t \le 9\,\mathrm{s}} \|e_p(t)\|_2 < 2\,\mathrm{mm}, \tag{4.11}$$

where $e_p(t)$ is the Cartesian position-error vector. This is an acceptance check that the commanded pose was retained, rather than a primary performance metric.

The redundant joint motion is evaluated from the projected joint velocity defined in Equation (2.59). The vector $\dot{q}_{\mathrm{null}} \in \mathbb{R}^7$ has units of rad/s and represents the component of the measured joint velocity that lies in the instantaneous kinematic null space.

Two complementary quantities are formed from this projected velocity. The first is the cumulative projected null-space motion

$$E_N = \int_{5\,\mathrm{s}}^{9\,\mathrm{s}} \|\dot{q}_{\mathrm{null}}(t)\|_2 \,\mathrm{d}t. \tag{4.12}$$

Because the norm is taken before integration, motion in opposite directions does not cancel. Thus E_N measures the total redundant joint motion and is particularly relevant to the damping comparison. Its unit is radians, and the reported values are converted to degrees.

The second quantity retains the direction of the motion. Integrating the projected velocity without taking its norm gives the accumulated projected joint displacement

$$\Delta q_{\mathrm{null}} = \int_{5\,\mathrm{s}}^{9\,\mathrm{s}} \dot{q}_{\mathrm{null}}(t)\,\mathrm{d}t = \int_{5\,\mathrm{s}}^{9\,\mathrm{s}} N_q(q(t))\dot{q}(t)\,\mathrm{d}t. \tag{4.13}$$

The vector $\Delta q_{\mathrm{null}} \in \mathbb{R}^7$ has units of radians. It describes the net accumulated projected joint motion during the disturbance interval, so motion in opposite directions can cancel. It is not an integrated torque: the integrated quantity is joint velocity.

A common direction is introduced to express this seven-dimensional net displacement by one signed scalar. Let $\Delta q_{\mathrm{null},0}$ denote the arithmetic mean of the three net projected displacement vectors recorded in the condition without null-space torque. The corresponding unit direction is

$$v_{\mathrm{ref}} = \frac{\Delta q_{\mathrm{null},0}}{\|\Delta q_{\mathrm{null},0}\|_2}. \tag{4.14}$$

Thus v_{ref} represents the redundant joint-space direction in which the commanded virtual disturbance moves the robot in the reference condition without a secondary null-space torque. The same direction is used for all tested null-space settings. Their signed net displacement along this common direction is

$$\Delta\eta_{\mathrm{dist}} = v_{\mathrm{ref}}^{\top}\Delta q_{\mathrm{null}}. \tag{4.15}$$

A positive $\Delta\eta_{\mathrm{dist}}$ denotes net motion in the same direction as the reference disturbance response, a negative value denotes net motion in the opposite direction, and a value near zero indicates little remaining net displacement along that direction. The quantity is a signed projection of the accumulated projected joint motion rather than an exact nonlinear redundant-coordinate displacement, because $N_q(q)$ changes with the robot configuration while v_{ref} is held fixed. At full rank the Panda has a one-dimensional instantaneous null space, so this common direction provides a meaningful signed comparison of the four tested settings.

The signed $\Delta\eta_{\mathrm{dist}}$ measures the net displacement remaining along the common disturbance direction and is especially relevant to the conditioning comparison. A trajectory can therefore have a large E_N but a small $\Delta\eta_{\mathrm{dist}}$. The change $\Delta\sigma_{\mathrm{min,dist}}$ is retained as a supporting controller diagnostic. The corresponding results are presented in Chapter 5.

Chapter 5

Results and Discussion

The surface-contact experiments evaluate how the controller parameters influence the end-effector rotation generated during Contact Establishment. The primary response quantity is the signed contact-establishment response γ_{t_i} about the investigated surface tangent t_i, defined in Section 4.5. It represents the end-of-contact-establishment orientation change relative to the orientation reference held through contact establishment. Its sign is chosen so that a response reducing the initial angular deviation carries the same sign as that deviation, and $|\gamma_{t_i}|$ is then the size of the correction.

The pose-based initial angular deviation θ_{0,t_i} defines the achieved experimental condition. The commanded tool orientation offset θ_{t_i} is the controller input that generates it, whereas γ_{t_i} is the measured response. The main response figures therefore identify each condition by θ_{0,t_i} and report γ_{t_i}. The corresponding numerical values are provided in Appendix D.

Each contact parameter setting was repeated three times. The comparative response values in Cases A–D are the arithmetic means over those repetitions, and the corresponding sample standard deviations are given in Appendix D. Repetition scatter is summarised once for each main case below. Within these cases, individual values are discussed only where their spread is relevant to the interpretation. The Case-D comparison additionally carries error bars. The preceding Tool Orientation state terminated with a finite tracking tolerance, so the initial angular magnitude differed slightly from the commanded value. At the TCP, the $+10°$ input produced mean initial deviations of $+9.31°$ about t_1 and $+10.05°$ about t_2. The reversed input produced $-9.41°$ and $-9.55°$, respectively.

5.1 Main Surface-Contact Results

The cases are reported in the order in which they constrain the choice of compliance-centre position. The TCP-centred reference is reported in Case A, followed by the stiffness sensitivities in Cases B and C. In Case D the tangential compliance-centre displacement was tested over both surface tangents and both signs of the pose-based initial angular deviation. The angular-magnitude, tool-axis, and intermediate-direction comparisons are reported as supporting checks in Appendix D.

5.1.1 Baseline Contact Response with the CoC at the TCP: Case A

In Case A the CoC was placed at the TCP, so $r_c = 0$. This condition removes the additional translation–rotation coupling introduced by the virtual point shift. The ordinary rotational impedance and physical contact geometry remain. The measured response therefore provides the baseline for the stiffness and compliance-centre-position comparisons. Repetition scatter remained at or below 0.19° across all Case-A settings in Table D.1.

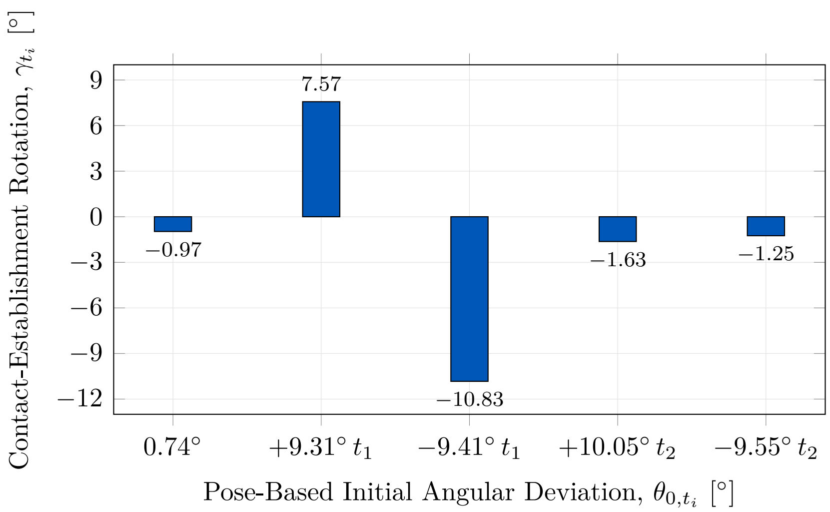


Figure 5.1: Measured contact-establishment rotation with the CoC at the TCP.

With the CoC at the TCP, the measured response differs strongly between the two surface tangents. For the positive initial-deviation conditions, Figure 5.1 gives +7.57° about t_1 and −1.63° about t_2. Against these positive initial deviations, the t_1 response was correction-directed, whereas the t_2 response had the opposite sign. Reversing the initial angular deviation changes the response to $-10.83°$ and $-1.25°$, respectively. The figure shows all five baseline conditions.

The nominal zero-offset input produced a pose-based initial deviation of 0.74° and gave $-0.97°$ about t_1. Contact produced end-effector rotation from the configured zero-offset target orientation as well. The other conditions are read against that non-zero response rather than against zero. The TCP-centred condition provides the reference against which the stiffness and compliance-centre-position variations are compared.

5.1.2 Effect of Cartesian Stiffness: Cases B and C

Rotational Stiffness: Case B

In Case B the rotational stiffness about the investigated tangent was varied while the other controller settings remained unchanged. The 5 N m/rad condition is the corresponding TCP-centred reference from Case A. Repetition scatter remained at or below 0.31° across all Case-B settings in Table D.2.+

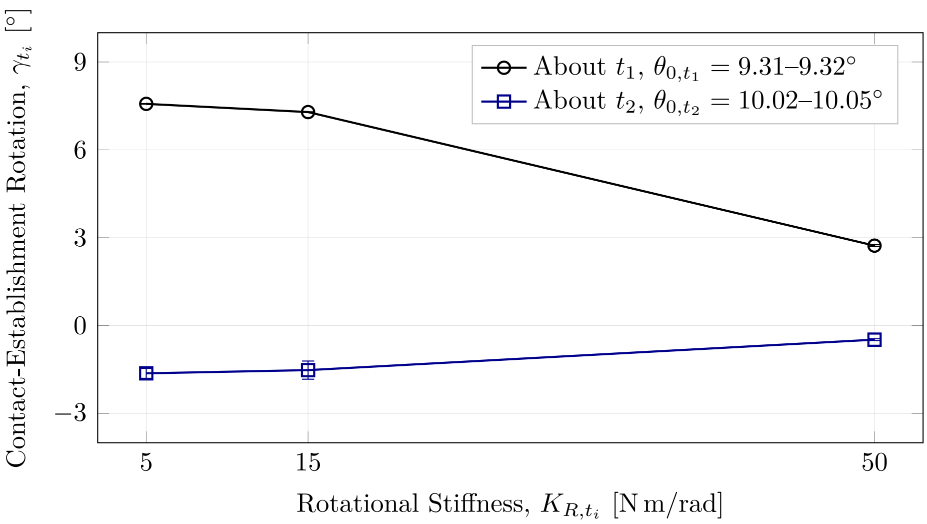


Figure 5.2: Measured contact-establishment rotation by rotational stiffness.

Increasing K_{R,t_i} reduced the measured rotation over the tested range. About t_1, Figure 5.2 shows a decrease from +7.57° at 5 N m/rad to +2.73° at 50 N m/rad. About t_2, the response remained small and changed from $-1.63°$ to $-0.48°$. Rotational stiffness therefore changed the response magnitude without removing the directional difference observed in Case A.

Cross-Axis Translational Stiffness: Case C

In Case C the translational stiffness perpendicular to the investigated rotation axis was varied. The rotational stiffness about that axis was held at 5 N m/rad. The 2000 N/m condition is the corresponding TCP-centred reference from Case A. Repetition scatter remained at or below 0.17° across all Case-C settings in Table D.3.

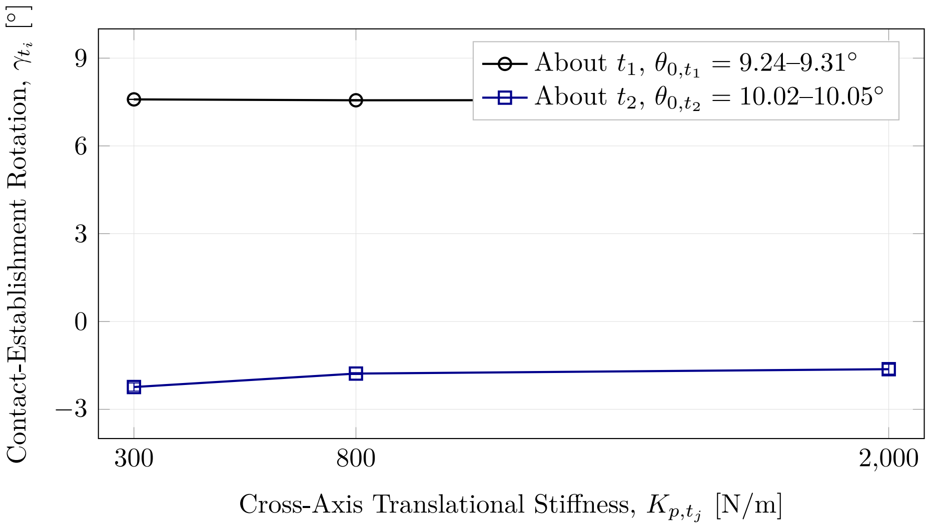


Figure 5.3: Measured contact-establishment rotation by cross-axis translational stiffness.

The influence of perpendicular translational stiffness was considerably smaller than that of rotational stiffness. About t_1, the measured rotation varied by only 0.03° across the tested range. About t_2, Figure 5.3 shows a change of 0.61°, from −2.24° to −1.63°. Varying this stiffness alone therefore did not account for the directional difference in the baseline response.

5.1.3 Effect of Tangential CoC Position: Case D

In Case D the tangential position of the CoC was varied while the Cartesian stiffness and damping remained unchanged. For each rotation axis, the centre was moved along the surface tangent perpendicular to that axis. The TCP-centred condition is included as the zero-position reference. Repetition scatter was small at most Case-D settings, and Table D.4 gives every sample standard deviation. Four conditions carry a larger spread, and they are named where the error bars appear below rather than summarised by a single bound.

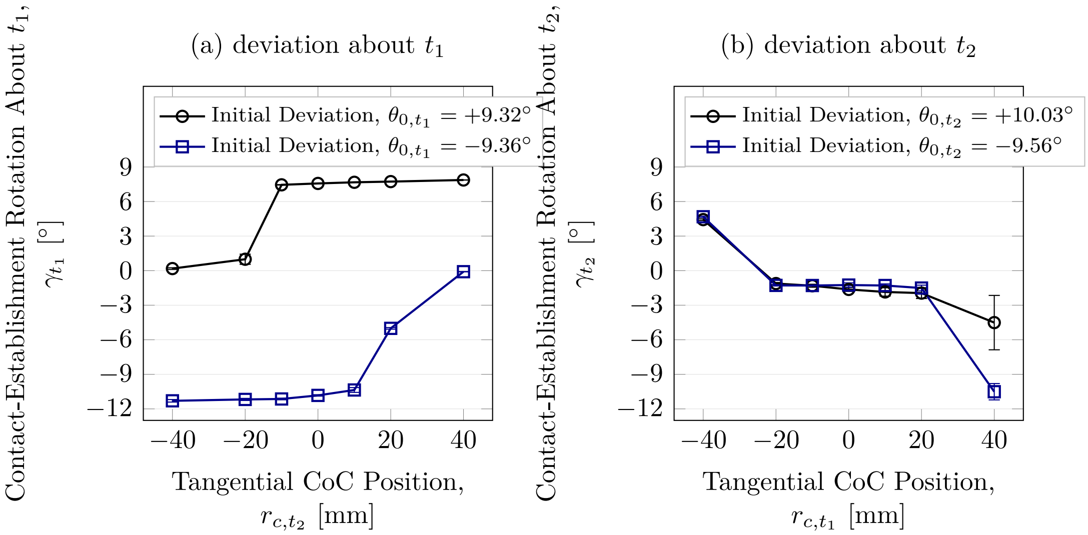


Figure 5.4: Contact-establishment rotation by tangential CoC position (mean ± SD).

As shown in Figure 5.4, reversing the initial orientation condition changes which side of the TCP produces the larger response. About t_1, the positive condition gives +7.87° at +40 mm and +0.18° at −40 mm. The reversed condition exchanges the larger response between the two sides.

Changing the rotation axis also changes the required tangential direction. For a positive condition about t_2, the response changes from −1.63° at the TCP to +4.43° at the selected $r_{c,t_1} = -40$ mm position. The selected displacement therefore lies along $+t_2$ for rotation about t_1 and along $-t_1$ for rotation about t_2. Among the tested positions, no non-zero tangential CoC was independent of the tangent direction and sign of the pose-based initial angular deviation.

The error bars in Figure 5.4 identify the conditions with the larger repetition spread. The positive t_1 condition at −20 mm had a standard deviation of 0.44°. About t_2, the positive condition had standard deviations of 0.47° at +20 mm and 2.36° at +40 mm. The reversed condition at +40 mm had a standard deviation of 0.71°.

The positive t_1 condition was examined in the time domain to relate the rotation response to the commanded wrench. Its two outer positions produce a large difference in γ_{t_1}, while the TCP-centred condition provides the zero-position reference. Each trace is one recorded repetition of its centre position.

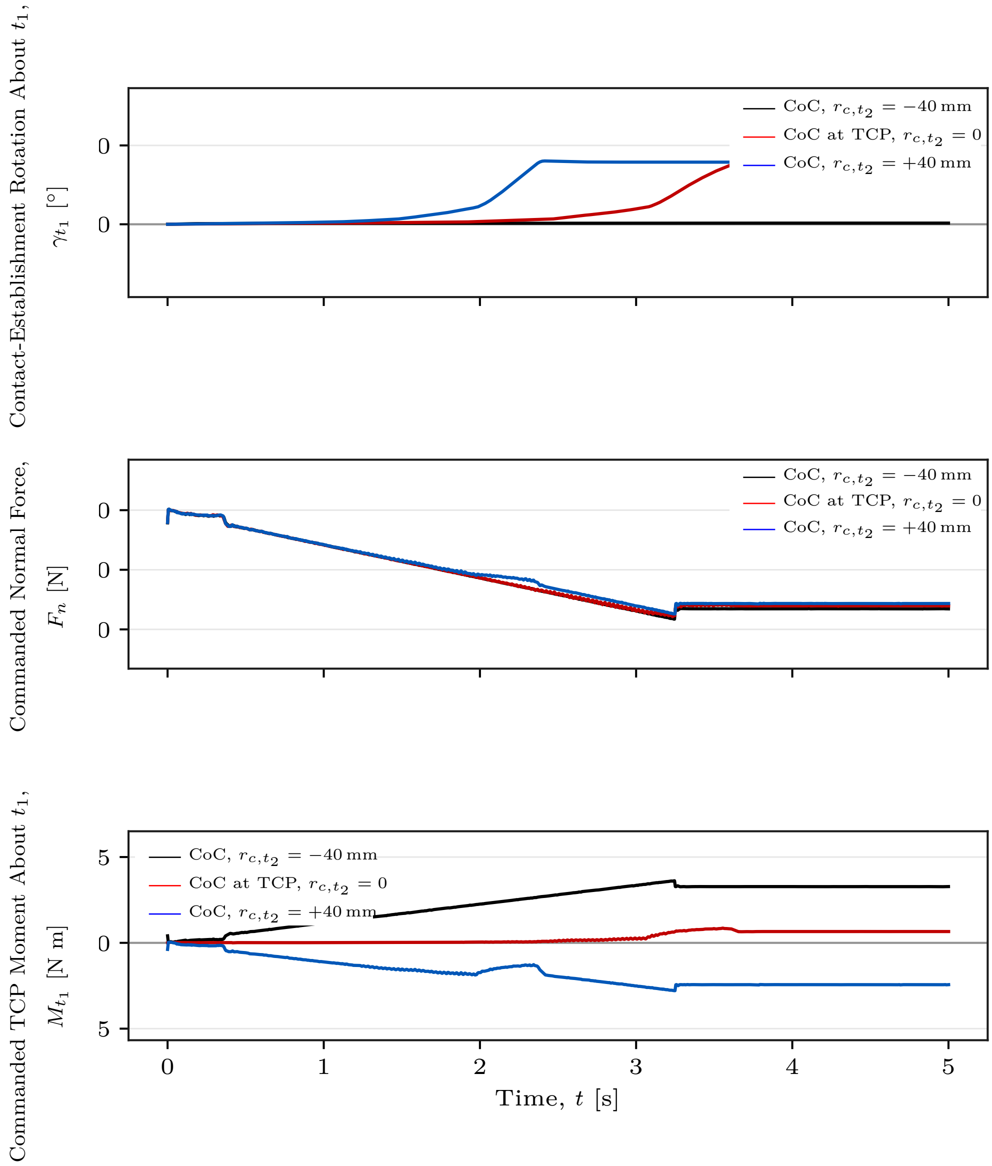


Figure 5.5: Contact-establishment response and commanded wrench at three CoC positions.

For this figure, the commanded Cartesian wrench $F = [f^\top, m^\top]^\top$ is resolved as

$$F_n = n_s^\top f, \qquad M_{t_1} = t_1^\top m.$$

The normal-force component is negative while the tool presses into the surface. Both quantities are controller commands.

Figure 5.5 places the contact-establishment rotation γ_{t_1}, the commanded normal force F_n, and the commanded TCP moment M_{t_1} on a common time axis for three centre positions. Every condition shown carries a commanded orientation offset of $+10°$ about t_1, corresponding to a mean pose-based initial deviation of approximately $+9.32°$. At -40, 0, and +40 mm, the commanded normal force settles in a similar range of approximately −77 to −82 N, while the commanded TCP moment about t_1 settles at approximately +3.28, +0.66, and −2.44 N m. The outer positions therefore produce moments of similar magnitude and opposite sign while their contact-establishment rotations differ by 7.73°. The difference in contact-establishment rotation is therefore associated primarily with the moment generated by the displaced impedance reference rather than with a substantial change in the normal pressing force.

At the selected +40 mm condition, the negative commanded moment M_{t_1} accompanies a positive γ_{t_1}. This response is formed from the measured end-effector orientation back to the reference held through contact establishment, as Section 4.5.2 defines it. Its sign is therefore opposite to the measured end-effector rotation from the start to the end of Contact Establishment.

A tangential displacement of the CoC moves the impedance reference point away from the TCP, so the translational press is coupled to the rotational part of the impedance. According to Equation (2.41), a non-zero tangential displacement adds the contribution $r_c \times f$ from the commanded press. This explains why the commanded moment changes strongly with centre position while F_n remains within a similar range. Reversing the tangential displacement reverses the contribution $r_c \times f_n$ generated by the normal press.

The measured sign reversal follows the relation derived in Section 2.7.2. A fixed tangential $r_c \neq 0$ retains one preferred moment direction while the press continues. At $r_c = 0$, the point-shift contribution vanishes and the TCP selects no tangential direction. The TCP-centred condition is therefore the direction-independent fixed reference among the tested positions. A displaced centre provides greater condition-specific rotational authority where the required direction is known.

The Grinding state returns to the decoupled impedance branch described in Section 3.2.7. This state structure is consistent with removing the direction-selective coupling after contact establishment. It is an implementation property rather than an experimental result, because the reported runs ended in the operator-controlled hold before grinding.

Across Cases A–D, tangential CoC position produced the largest measured response variation among the tested controller parameters.

5.2 Null-Space Pose-Hold Results

The remaining experiment is separate from the contact cases and analyses the redundant joint-space direction while the Cartesian pose is held. Matched trials compare the selectable null-space terms under the same internally commanded point-force-equivalent disturbance. The commanded disturbance reached 20.00 N in all twelve trials, with peak equivalent joint-torque norms between 2.17 and 2.45 N m. The torque-limit scale remained equal to one, so the waveform was applied without clipping.

Figure 5.6 reports the response during the Cartesian pose-hold disturbance test in three panels. Panel (a) gives the mean cumulative projected null-space motion E_N without null-space torque and with projected damping, the shaded bands showing $\pm$ one standard deviation across the three repetitions. Panel (b) gives the change in the minimum Jacobian singular value over the disturbance interval, $\Delta\sigma_{\mathrm{min,dist}}$, together with the peak Cartesian position error, for the two singular-value conditioning magnitudes; its error bars show $\pm$ one standard deviation and the horizontal line marks the 2 mm position-retention limit. Panel (c) gives the net redundant displacement $\Delta\eta_{\mathrm{dist}}$ of all four settings, with the mean printed above each bar and error bars showing $\pm$ one standard deviation. Selected numerical values are discussed below.

5.2.1 Projected Null-Space Damping

Projected null-space damping reduced E_N from 7.596° without null-space torque to 5.687° at $d_{\mathrm{null}} = 2\,\mathrm{N\,m\,s/rad}$, a reduction of approximately 25 %; panel (a) of Figure 5.6 carries both traces. The net redundant displacement decreased in the same proportion, from 0.131 rad to 0.098 rad, while the change in σ_{min} across the disturbance interval became less negative, from -2.13×10^{-3} to -1.26×10^{-3}. This behaviour follows from the projected damping law of Equation (2.61). The torque vanishes as the redundant motion stops and therefore requests no return towards the initial configuration.

5.2.2 Singular-Value Conditioning

Singular-value conditioning behaved differently from both the uncontrolled and the damping-only settings. It was already active during the 5 s preceding the disturbance and

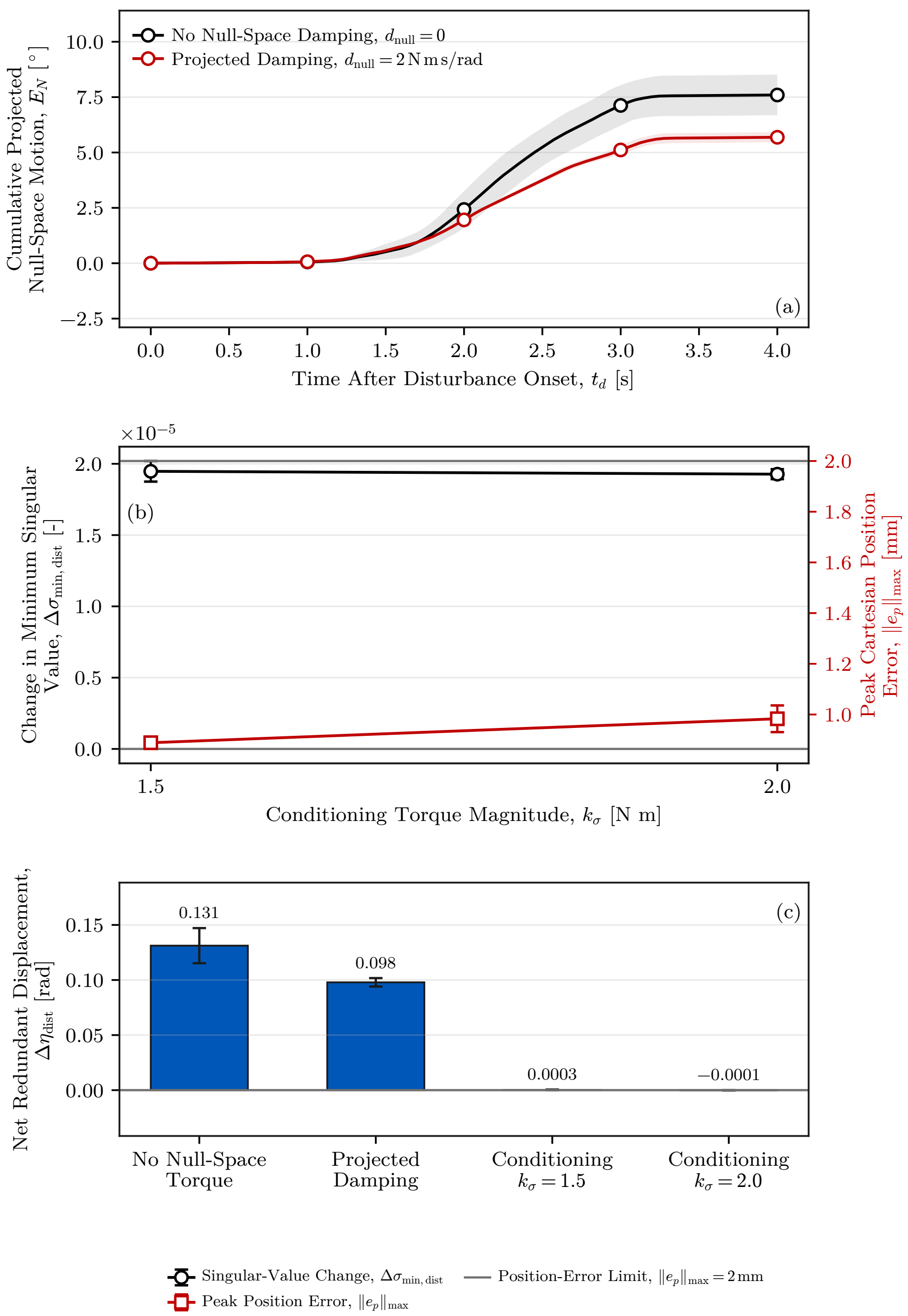


Figure 5.6: Null-space pose-hold response.

remained active throughout it, so it did not act as a term applied after the redundant configuration had been displaced. Both sigma-only settings ended the disturbance interval with a net redundant displacement close to zero: 0.3×10^{-3} rad at $k_\sigma = 1.5\,\mathrm{N\,m}$ and -0.1×10^{-3} rad at $2.0\,\mathrm{N\,m}$. As panel (c) of Figure 5.6 shows, both values are more than two orders of magnitude smaller than the 0.131 rad obtained without null-space torque. Over the same interval $\sigma_{\min}$ increased by approximately 1.95×10^{-5} and 1.93×10^{-5}, whereas it decreased by 2.13×10^{-3} without null-space torque. Under the tested disturbance direction, the conditioning term therefore strongly suppressed the resulting redundant displacement.

The two sigma magnitudes differed mainly in motion activity. At $k_\sigma = 2.0\,\mathrm{N\,m}$, the cumulative projected null-space motion was 1.619°, compared with 0.283° at 1.5 N m, a factor of about six, while both net displacements remained near zero. The larger torque magnitude was associated with greater switching activity near the locally preferred region. This is consistent with the control law, because the conditioning term commands a fixed torque magnitude along the selected direction and the selector s_σ becomes zero only inside the comparison deadband. The larger cumulative motion therefore indicates greater switching activity around the same region rather than poorer disturbance rejection. The mean peak Cartesian position error was 0.889 mm at 1.5 N m and 0.983 mm at 2.0 N m. The largest position error measured in the complete pose-hold study was 1.304 mm, below the predefined 2 mm acceptance limit. Of the two tested magnitudes, 1.5 N m achieved comparable suppression of net disturbance-induced displacement with substantially less redundant motion.

5.2.3 Interpretation and Experimental Bounds

The pose-hold trials compare the four tested null-space settings under the repeatable commanded disturbance. The two terms act on different quantities. Projected damping dissipates redundant joint motion that is already present, whereas the singular-value term drives the redundant configuration towards the sampled neighbour with the larger $\sigma_{\min}$ whether or not such motion is present. The conditioning law is not dissipative: a disturbance acting towards larger $\sigma_{\min}$ would move the configuration in the same direction as τ_σ rather than against it, so the rejection observed here is specific to the tested disturbance direction. Both terms act through the null-space projector, so neither contributes a first-order Cartesian wrench.

Several bounds apply to these results. Because the conditioning torque was active before the disturbance, the sigma-only trials entered it from a redundant configuration displaced by approximately 0.013 rad from the one at which the trials without conditioning began. The four settings therefore compare the closed-loop behaviour of the complete modes. They do not isolate an instantaneous opposing torque at identical joint configurations. The surface-contact experiments use both contributions together in the combined secondary-torque mode and therefore do not separate their individual effects during contact. The geometric Jacobian combines translational and rotational rows without a common length scale, so the reported $\sigma_{\min}$ is used only as a relative indicator under the fixed Jacobian convention.

Chapter 6

Conclusion and Future Work

6.1 Conclusion

A real-time Cartesian impedance controller was implemented for contact-induced alignment of a rectangular tool with a configured surface reference. The controller assigns directional stiffness and damping in the surface frame and can shift the virtual centre of compliance from the TCP. The shift couples the commanded press to the rotational impedance through the term $r_c \times f$.

The TCP-centred conditions established a direction-dependent contact response. Substantial alignment-directed end-effector rotation was measured about t_1, while the response about t_2 was smaller and had the opposite sign for the positive command. Raising the rotational stiffness reduced the response over the tested range. Changing the cross-axis translational stiffness produced a smaller effect.

Tangential compliance-centre position produced the largest measured variation. The displacement that increased alignment depended on the sign and axis of the pose-based initial angular deviation. Reversing the displacement reversed the moment contribution from the normal press. Within the investigated contact configuration and tested positions, a non-zero tangential centre could therefore not be selected independently of the direction and sign of that initial deviation.

The TCP-centred impedance is the direction-independent fixed centre of compliance among the tested positions, because it selects no tangential direction. A displaced centre provides greater condition-specific rotational authority when the required direction is known. These properties are distinct: the fixed reference need not produce the largest alignment-directed rotation in every condition.

The null-space pose-hold study separated projected joint damping from singular-value conditioning. Damping reduced redundant joint motion while the disturbance acted. Under the tested disturbance direction, conditioning strongly suppressed the net redundant displacement and kept the Cartesian position error below the specified 2 mm limit. The lower tested conditioning magnitude produced the same suppression with less cumulative redundant motion.

6.2 Limitations

6.2.1 Experimental Range

The contact study used one Panda configuration, one tool and mounting arrangement, one configured surface reference, and one press trajectory. The conclusions are therefore limited to this geometry and the tested stiffness, initial angular-deviation and compliance-centre ranges. Sustained grinding and a centre scheduled during contact were not evaluated.

6.2.2 Measurement and Tool-Mount Uncertainty

The experimental condition was the pose-based initial angular deviation from the configured surface reference. The physical plane normal was not measured independently. The primary response was calculated from the measured end-effector orientations. The tool mount exhibited approximately $\pm 2^\circ$ of unloaded play about y_{EE}, and its instantaneous motion was not tracked during contact. The results therefore describe end-effector rotation and do not provide an independent physical tool–surface angle.

6.2.3 Null-Space Configuration and Real-Time Operation

The null-space terms were separated only in Cartesian pose hold under one internally commanded disturbance direction. The conditioning torque was active before the disturbance, so the experiment compares complete closed-loop modes rather than isolated torque contributions at identical configurations. The surface-contact study held the combined mode fixed. Worst-case execution time and scheduling jitter were not measured separately from operation through the nominal 1 kHz interface.

6.3 Future Work

The first priority is an independent measurement of the physical plane normal and the physical tool-face angle during contact. This would quantify the surface-reference error and separate end-effector rotation from motion in the tool mount.

The second priority is a direction-selected compliance-centre strategy. Such a controller would begin at the TCP, estimate the required alignment direction, apply a bounded tangential displacement during contact establishment, and return to the TCP before sustained contact. Negative orientation offsets and additional tangent-plane directions should be included in its evaluation.

The third priority is a combined-mode null-space study under physical disturbance. Varying projected damping and conditioning together would show whether the pose-hold findings remain valid during contact. A continuous or hysteretic conditioning law should also be compared with the present two-point selector to reduce its switching activity.

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Appendix A

Controller Source Listings

ORANGE — sufficient. The listings are abridged to the mathematical paths used by the thesis and connect each important equation to an implementation excerpt.

The following listings show the implementation of the compliance-centre transformation and the final torque assembly. They use fixed-size Eigen aliases such as `Vec3`, `Vec7`, `Mat3`, `Mat6x6` and `Mat6x7`, and `R_base_surface` denotes the surface-frame orientation used in the chapters.

A.1 Centre of Compliance and the Offset Adjoint

The thesis displacement $r_c = p_c - p_{\mathrm{TCP}}$ points from the TCP to the selected centre of compliance. The controller source evaluates the same transform with the internal lever $\ell_c = p_{\mathrm{TCP}} - p_c = -r_c$. The source identifier `r_c` in the following listings denotes ℓ_c, not the thesis quantity r_c. Consequently, $[\ell_c]_\times = -[r_c]_\times$, and the two displayed adjoints are identical. The same congruence is applied to stiffness and damping.

Listing A.1: Source-side compliance-centre adjoint and congruence transform.

```
Mat3 skewMatrix(const Vec3& v) {
  Mat3 s;
  s << 0.0, -v(2),  v(1),
       v(2),  0.0, -v(0),
      -v(1),  v(0),  0.0;
  return s;
}

Mat6x6 complianceShiftMatrix(const Vec3& r_c) {
  Mat6x6 adjoint = Mat6x6::Zero();
  adjoint.block<3, 3>(0, 0) = Mat3::Identity();
  adjoint.block<3, 3>(0, 3) = skewMatrix(r_c);
  adjoint.block<3, 3>(3, 3) = Mat3::Identity();
  return adjoint;
}
```

```

Mat6x6 shiftGainToTcp(
    const Mat6x6& gain_at_center, const Vec3& r_c) {
  const Mat6x6 adjoint = complianceShiftMatrix(r_c);
  return adjoint.transpose() * gain_at_center * adjoint;
}
```

The tool-frame parameter `compliance_center_offset_ee` stores the thesis displacement r_c and is negated when the internal lever is formed. The alternative surface-frame parameter `r_tcp_from_compliance_center_surface` stores ℓ_c directly. Only the tool-frame definition was used for the reported non-zero displacements.

Listing A.2: Selection of the decoupled or coupled spring wrench (abridged).

```
const bool coupled =
    params.use_virtual_compliance_center &&
    (state == ControlState::kContactEstablishment ||
     state == ControlState::kPreGrindingHold ||
     (state == ControlState::kPoseHold &&
      params.use_contact_impedance_hold));
if (!coupled) {
  // Two independent springs. dv.tail is -omega, so the damping
  // signs match.
  Vec6 wrench;
  wrench.head<3>() = Kp * dx.head<3>() + Dp * dv.head<3>();
  wrench.tail<3>() = KR * dx.tail<3>() + DR * dv.tail<3>();
  return wrench;
}

Vec3 r_c = Vec3::Zero();
if (params.compliance_center_in_tool_frame) {
  r_c = -(R_EE * params.compliance_center_offset_ee);
} else {
  r_c = R_base_surface *
      params.r_tcp_from_compliance_center_surface;
}
const Mat6x6 K_tcp =
    shiftGainToTcp(blockDiagonal(Kp, KR), r_c);
const Mat6x6 D_tcp =
    shiftGainToTcp(blockDiagonal(Dp, DR), r_c);
return K_tcp * dx + D_tcp * dv;
```

A.2 Core Control Callback

The two listings below show the state and model quantities acquired at the start of each torque callback and how they enter the torque command.

Listing A.3: Robot-state and model quantities acquired by the torque callback.

```
Map<const Vec7> q_current(state.q.data());
Map<const Vec7> dq(state.dq.data());

const auto jacobian_array =
    model.zeroJacobian(Frame::kEndEffector, state);
Map<const Mat6x7> J(jacobian_array.data());
const Vec6 xdot = J * dq;

Map<const Mat4x4> T_EE(state.O_T_EE.data());
const Vec3 p_EE = T_EE.block<3, 1>(0, 3);
const Mat3 R_EE = T_EE.block<3, 3>(0, 0);
```

Listing A.4: Core impedance, coupled contact establishment, and torque composition.

```
const Vec3 e_p = desired.p_d - p_EE;
const Vec3 e_R = applyRotationalAxisMask(
    params, orientationError(R_EE, R_d_used), R_base_surface);

Vec6 dx;
dx.head<3>() = e_p;
dx.tail<3>() = e_R;
Vec6 dv;
dv.head<3>() = desired.pdot_d - pdot;
dv.tail<3>() = -omega;

const Vec6 wrench =
    computeCartesianImpedanceWrench(
        params, state, Kp_used, Dp_used,
        KR_used, DR_used, R_base_surface,
        dx, dv, R_EE);

const Vec7 tau_nullspace = computeNullspaceTorque(
    params, model, state, J, dq);

AutomaticDisturbance disturbance;
if (state == ControlState::kPoseHold &&
    params.disturbance_auto_enabled) {
  disturbance = computeAutomaticDisturbance(
      params, model, state, disturbance_force_direction_base,
      time - state_start_time);
}

Array7 coriolis_array = model.coriolis(state);
Map<const Vec7> coriolis(coriolis_array.data());
const Vec7 tau_task = J.transpose() * wrench;
const Vec7 tau_cmd =
    tau_task + tau_nullspace + disturbance.tau + coriolis;
```

The disturbance term is non-zero only in the Cartesian pose-hold study of Section 4.6 and is zero in every surface-contact run.

Appendix B

Recorded Experimental Data

ORANGE — sufficient. The appendix records the signals actually used in the evaluation and identifies the common configured surface reference used by the controller and analysis.

The controller stores experimental signals in a preallocated ring buffer. This avoids file access and dynamic buffer growth inside the 1 kHz control callback. After the controller stops, the retained rows are written in chronological order to a CSV file.

B.1 Control States

ORANGE — sufficient. The numeric phase values are mapped directly to their recorded state names.

The source field `state` identifies the active controller state:

0	`tool_orientation`	7	`pre_grinding_hold`
1	`surface_approach`	3	`grinding`
6	`pre_contact_hold`	4	`cartesian_pose_hold`
2	`contact_establishment`	5	`manual_guidance`

B.2 Recorded Signals

ORANGE — sufficient. The table is a useful reproducibility record and keeps units, frames, and model-estimated quantities explicit.

Table B.1 lists the union of two CSV schemas used for the reported experiments. No individual file contains every group in the table. The surface-contact logs contain the contact-geometry fields and use the current angular-deviation names. The archived Cartesian pose-hold logs contain the singular-value probe and projected-motion fields and retain the earlier alignment-error names. Command, state, wrench, and disturbance groups common to the two schemas are listed once. A suffix `_x,_y,_z` denotes base-frame components, and `_1.._7` denotes joint components.

Table B.1: Recorded signal groups.

Column group	Meaning
`time`, `state`	Run time and active control state.
`p_EE`, `p_d`	Measured and desired TCP positions [m].
`tool_contact`, `edge_target`	Selected tool reference point and its target [m].
`tool_contact_offset_ee`	Offset of the selected tool reference point from the TCP, in EE coordinates [m].
`first_contact_tcp`, `first_contact`	TCP position and projected surface reference captured at the clearance transition [m].
`e_p`, `e_R`, `pdot`, `pdot_d`, `omega`	Cartesian errors and velocities; during Contact Establishment `e_R` is referred to the orientation held from the clearance transition.
`angular_deviation_deg`	Pose-based alignment error $\theta_{\text{align}} = \cos^{-1}(-n_{\text{Tool},0}^{\top} n_s)$, used in the consistency check of Appendix D.6 [°]. The archived pose-hold schema names this field `alignment_angle_deg`.
`angular_deviation_t1_deg`, `angular_deviation_t2_deg`, `angular_deviation_normal_deg`	The same angular deviation resolved on t_1, t_2 and n_s [°]. The archived pose-hold aliases begin with `alignment_error_`. These are not evaluation quantities.
`p_CoC`, `r_eff`	CoC $p_{\text{TCP}} + r_c$ and the offset from it to the selected tool point, $p_{\text{Tool}} - p_c$, in the base frame [m]. Written for each surface-contact run, including $r_c = 0$.

Table B.1 continued

Column group	Meaning
`t_align`	Time from the start of Contact Establishment at which the deviation first settled inside the configured tolerance [s], negative until then. Observed only; it does not end the state.
`f`, `m`	Commanded Cartesian force [N] and moment [N m].
`external_force`, `external_moment`	Model-estimated external wrench from libfranka, expressed in the base orientation and referenced to stiffness frame K.
`contact_force_bias`, `contact_moment_bias`	External-wrench values captured at the clearance transition.
`force_after_contact`, `moment_after_contact`	The model-estimated wrench with those reference values already subtracted.
`push`	Contact-establishment press coordinate or grinding preload reference retained afterwards [m].
`nullspace_mode`	Selected null-space law.
`sigma_current`	Current smallest singular value $\sigma_{\min}$ in the archived pose-hold schema.
`sigma_plus`, `sigma_minus`, `sigma_difference`, `sigma_direction`	$\sigma_{\min}$ at the two sampled configurations $q \pm \alpha_{\text{probe}} v_7$, their difference and the selected sign. These fields belong to the archived pose-hold schema.
`nullspace_dq_1..7`	Projected joint velocity $\dot{q}_{\text{null}}$ [rad/s].
`tau_sigma_1..7`, `tau_sigma_norm`, `tau_nullspace_norm`	Singular-value torque τ_σ and total null-space torque τ_{null} [N m].
`nullspace_damping`	Active projected damping coefficient [N m s/rad].

Table B.1 continued

Column group	Meaning
`disturbance_scale`, `disturbance_torque_scale`, `disturbance_force_base_x..z`, `disturbance_point_base_x..z`	Commanded hold-disturbance waveform, torque-limit scale, point force [N], and the base-frame position of the link point it acts at [m].
`tau_disturbance_1..7`	Equivalent joint torque $J_p^\top f_d$ in the internally commanded hold study [N m].
`tau_cmd_1..7`	Final commanded joint torque [N m].

Appendix C

Controller Parameters

ORANGE — sufficient. The active geometry, phase gains, damping policy, timings, null-space values, and collision settings are collected in one reproducibility record.

This appendix lists the effective controller parameters used in the reported experiments. It is a reproducibility record for the implementation described in Chapters 3 and 4.

C.1 Surface and Tool Geometry

ORANGE — sufficient. The configured geometry and initial joint vector are stated with their exact keys and units.

The plane satisfies

$$n_s^\top (p - p_s) = 0, \qquad n_s = R_y(\beta_s) R_x(\alpha_s) e_{z_0},$$

and the surface-frame column order is $[t_1, t_2, n_s]$. Table C.1 collects the corresponding surface and tool parameters.

Table C.1: Surface and tool parameters.

Quantity	Configuration key	Value
Surface point	`surface_point`	$[0.515267, -0.107172, 0.003108]\,\mathrm{m}$
Surface tilts	α_s, β_s	-1.585°, $+0.988^\circ$
First-tangent hint	`surface_tangent1_hint_base`	$[1, 0, 0]$
Tool axis in EE	`tool_axis_ee`	$[0.026387057, -0.006678514, 0.999629492]$
Signed alignment target	`tool_axis_target_sign`	$-n_s$
Spin about the tool axis, from t_1	`tool_target_offset_normal_deg`	90.0°
Tool-face centre in EE	`tool_contact_face_center_ee`	$[0, 0, 0.020]\,\mathrm{m}$
Tool face size	width $\times$ length	$0.040 \times 0.120\,\mathrm{m}$
Tool-point selection tolerance, $\varepsilon_{\mathrm{sel}}$	`tool_contact_feature_tie_tolerance`	$0.1\,\mathrm{mm}$

C.1.1 Calibration Relations

For the seated surface pose, the nominal end-effector axis and the tool-face centre offset supplied the stored plane quantities,

$$n_s = R_{\mathrm{EE}}[0, 0, 1]^\top, \qquad p_s = p_{\mathrm{EE}} + R_{\mathrm{EE}} r_{\mathrm{face,EE}}. \tag{C.1}$$

The first tangent was the normalised projection of the base-frame x direction onto the plane, and $t_2 = n_s \times t_1$ completed the surface frame. For the tool-normal calibration, the fitted direction was

$$n_{\mathrm{Tool,EE}} = \arg\min_{\|n\|=1} \sum_{i<j} \|(R_{\mathrm{cal},i} - R_{\mathrm{cal},j})n\|^2 . \tag{C.2}$$

Three seated yaw captures entered the fit. The sign of the fitted eigenvector was selected towards nominal $+Z_{\mathrm{EE}}$.

C.1.2 Initial Joint Configuration

The initial joint configuration used in the reported experiments is

$$q_{\text{init}} = [0.014777,\ 0.210911,\ -0.009320,\ -2.284835,\\ -0.002725,\ 2.493410,\ 0.780694]^\top \text{ rad}.$$

C.2 State Sequence and Impedance

ORANGE — sufficient. The phase-specific gain frames and transition parameters reconcile the values stated across Chapters 3 and 4.

Table C.2: State and configuration stiffness and damping.

State / quantity	Active stiffness and frame	Configured damping
Approach translation	$[150, 150, 150]\,\text{N/m}$, surface $[t_1, t_2, n_s]$	$[20, 20, 20]\,\text{N s/m}$
Approach rotation	$[180, 180, 90]\,\text{N m/rad}$, surface $[t_1, t_2, n_s]$	$[15, 15, 12]\,\text{N m s/rad}$
Contact Establishment translation	$[2000, 2000, 350]\,\text{N/m}$, surface $[t_1, t_2, n_s]$	$[10, 10, 175]\,\text{N s/m}$
Contact-Establishment rotation	$[5, 5, 50]\,\text{N m/rad}$, surface $[t_1, t_2, n_s]$	$[10.01, 10.01, 10]\,\text{N m s/rad}$
Hold translation	$2000\,\text{N/m}$, isotropic base	$50\,\text{N s/m}$
Hold rotation	$50\,\text{N m/rad}$, isotropic base	$8\,\text{N m s/rad}$
Manual guidance	no Cartesian stiffness; joint coordinates	$0.5\,\text{N m s/rad}$ joint damping

The active timing and trajectory values are listed in Table C.3.

Table C.3: State-transition and trajectory parameters.

Quantity	Configuration key	Value
Orient minimum time	`approach_orient_min_time`	1.0 s
Tool-axis switch tolerance	`approach_orient_error_threshold`	0.016 rad
Spin switch tolerance	`approach_orient_spin_error_threshold`	0.009 rad
Orient timeout / maximum rate	`approach_orient_timeout`, `approach_orient_max_rate_deg`	8.0 s, 20°/s
Descent speed / maximum distance	`descend_speed`, `descend_max_distance`	0.1 m/s, 0.640 m
Tool clearance, $h_{\text{clearance}}$	`descend_surface_clearance`	0.020 m
Contact-establishment minimum time / timeout	`contact_establishment_min_time`, `contact_establishment_timeout`	2.0 s, 5.0 s
Contact-establishment moment-change threshold	`contact_establishment_moment_threshold`	150 N m
Contact-establishment push speed / endpoint	`contact_establishment_push_speed`, `contact_establishment_push_end`	0.080 m/s, 0.240 m in all surface-contact experiments
Pre-contact hold	`enable_pre_contact_hold`	disabled
Pre-grinding hold	`enable_pre_grinding_hold`	enabled

C.3 Inertia-Scaled Damping and Centre of Compliance

Table C.4: Damping and compliance-centre parameters.

Quantity	Symbol / key	Value
Approach damping factor	ζ	1.9
Contact-establishment damping factor	ζ	1.0
Operational-space-inertia regularisation	added to $J(q)M(q)^{-1}J^{\top}(q)$	10^{-9}
Maximum damping coefficient	`auto_damping_max`	400
Lower bound from the configured matrices	`auto_damping_min_from_manual`	0; those matrices act as fallback only
Reported contact states using inertia-scaled damping	—	Surface Approach and Contact Establishment
Pose-hold inertia-scaled damping	`hold_auto_damping`	0 in all twelve reported runs
Pose-hold translational damping	D_p	$50I_3\,\mathrm{N\,s/m}$
Pose-hold rotational damping	D_R	$8I_3\,\mathrm{N\,m\,s/rad}$
Coupled 6×6 law	`use_virtual_compliance_center`	Contact Establishment only, and only for displaced-centre conditions
Displacement definition frame	`compliance_center_in_tool_frame`	tool-fixed in all reported experiments
Tool-fixed displacement	`compliance_center_offset_ee`	in end-effector axes

The reported displacement defines $r_c = p_c - p_{\mathrm{TCP}}$ in end-effector coordinates. The damping calculation and the compliance-centre transformation are described in

Sections 2.6.1 and 2.7, and the tested displacement components are listed in Table 4.4 and Appendix D.

C.4 Null-Space and Operating Parameters

ORANGE — sufficient. The selected mode, gains, probe settings, logging capacity, gripper state, and robot-side thresholds are stated explicitly.

The null-space settings are explained in Section 4.3.2. The final two rows of Table C.5 are Franka collision/reflex thresholds configured through the libfranka collision-behaviour interface. Robot-side monitoring applied these thresholds throughout the reported runs.

Table C.5: Null-space and operating parameters.

Parameter	Symbol / key	Value
Null-space mode	`nullspace_mode`	3 (projected damping and singular-value conditioning)
Null-space damping	d_{null}	2.0 N m s/rad
Sigma torque magnitude	k_σ	1.0 N m
Probe distance	α_{probe}	0.08 rad
Sigma difference deadband	δ_σ	10^{-6}
Relative SVD cutoff	ρ	10^{-4}
Recorded-data sampling / capacity	`log_every_n_cycles`, rows	every callback (1), 120000
Run duration / stop	`experiment_duration`	0 s: indefinite until `e+Enter`
Gripper grasp	width, speed, force	0.066 m, 0.050 m/s, 60 N
Tool-mount rotational play	observed pad-mount clearance	approximately $\pm 2^\circ$ about y_{EE}
Manual-guidance damping	`manual_guidance_damping`	0.5 N m s/rad
Joint-side collision threshold	acceleration/nominal	140 N m on each axis
Cartesian collision threshold	acceleration/nominal	240 N or N m on each channel

Appendix D

Supporting Contact Checks and Results

This appendix reports the complete numerical values behind the main contact figures and three supporting checks. The supporting checks comprise 45 runs across 15 independently counted settings, with three repetitions per setting. The calibrated geometry, state sequence, Cartesian stiffness and damping, null-space settings, and evaluation quantities were those of Sections 4.2 and 4.3 unless a changed condition is stated below.

D.1 Numerical Values for Main Cases A–D

The main chapter retains the comparison figures and interpretation. The tables below provide the corresponding numerical values, each as the arithmetic mean and sample standard deviation over the three repetitions of that setting. In Table D.4 the reported coordinate is r_{c,t_2} for initial deviations about t_1 and r_{c,t_1} for initial deviations about t_2, so the displacement selected for a positive deviation appears at $+40\,\mathrm{mm}$ in the upper table and at $-40\,\mathrm{mm}$ in the lower one.

Table D.1: Case-A baseline response at the TCP, mean ± SD.

Pose-Based Initial Angular Deviation	CoC Position	Measured Contact-Establishment Rotation γ_{t_i} [°]
0.74° magnitude	TCP	-0.97 ± 0.02
+9.31° about t_1	TCP	$+7.57 \pm 0.01$
−9.41° about t_1	TCP	-10.83 ± 0.06
+10.05° about t_2	TCP	-1.63 ± 0.17
−9.55° about t_2	TCP	-1.25 ± 0.19

Table D.2: Case-B rotational-stiffness response, mean ± SD.

Pose-Based Initial Angular Deviation	CoC Position	Varied Rotational Stiffness K_{R,t_i} [N m/rad]	Reference	Measured Contact-Establishment Rotation γ_{t_i} [°]
+9.31° about t_1	TCP	5	Case-A reference	$+7.57 \pm 0.01$
+9.32° about t_1		15	—	$+7.29 \pm 0.01$
+9.31° about t_1		50	—	$+2.73 \pm 0.03$
+10.05° about t_2	TCP	5	Case-A reference	-1.63 ± 0.17
+10.05° about t_2		15	—	-1.52 ± 0.31
+10.02° about t_2		50	—	-0.48 ± 0.03

Table D.3: Case-C cross-axis translational-stiffness response, mean ± SD.

Pose-Based Initial Angular Deviation	CoC Position	Varied K_{p,t_j}	K_p [N/m]	Reference	Measured Contact-Establishment Rotation γ_{t_i} [°]
+9.27° about t_1	TCP	K_{p,t_2}	300	—	$+7.59 \pm 0.00$
+9.24° about t_1		K_{p,t_2}	800	—	$+7.56 \pm 0.02$
+9.31° about t_1		K_{p,t_2}	2000	Case-A reference	$+7.57 \pm 0.01$
+10.04° about t_2	TCP	K_{p,t_1}	300	—	-2.24 ± 0.14
+10.02° about t_2		K_{p,t_1}	800	—	-1.78 ± 0.03
+10.05° about t_2		K_{p,t_1}	2000	Case-A reference	-1.63 ± 0.17

Table D.4: Case-D tangential compliance-centre response, mean ± SD.

Initial deviation about t_1; coordinate r_{c,t_2}							
Deviation	-40	-20	-10	**TCP**	$+10$	$+20$	$+40$ mm
$+9.32°$	0.18 ± 0.07	0.99 ± 0.44	7.45 ± 0.02	7.57 ± 0.01	7.67 ± 0.01	7.73 ± 0.01	7.87 ± 0.01
$-9.36°$	-11.30 ± 0.11	-11.18 ± 0.09	-11.14 ± 0.07	-10.83 ± 0.06	-10.36 ± 0.21	-5.01 ± 0.07	-0.09 ± 0.01

Initial deviation about t_2; coordinate r_{c,t_1}							
Deviation	-40	-20	-10	**TCP**	$+10$	$+20$	$+40$ mm
$+10.03°$	4.43 ± 0.07	-1.12 ± 0.09	-1.31 ± 0.13	-1.63 ± 0.17	-1.85 ± 0.36	-1.95 ± 0.47	-4.51 ± 2.36
$-9.56°$	4.68 ± 0.05	-1.28 ± 0.05	-1.29 ± 0.01	-1.25 ± 0.19	-1.29 ± 0.14	-1.51 ± 0.25	-10.51 ± 0.71

D.2 Initial Angular-Deviation Magnitude

The 5° and 10° controller inputs generated two pose-based initial angular deviations about each tangent. They were compared with the CoC at the TCP and at the selected 40 mm tangential position. These four independently counted settings gave 12 runs.

Table D.5: Initial-deviation response at two compliance-centre positions.

Rotation Direction	Pose-Based Initial Deviation	TCP-Centred γ_{t_i} [°]	Selected 40 mm CoC γ_{t_i} [°]	Effect of the CoC [°]
About t_1	4.28°	+2.55	+2.70	+0.15
About t_1	9.32°	+7.57	+7.87	+0.30
About t_2	5.16°	−1.40	+5.91	+7.31
About t_2	10.03°	−1.63	+4.43	+6.06

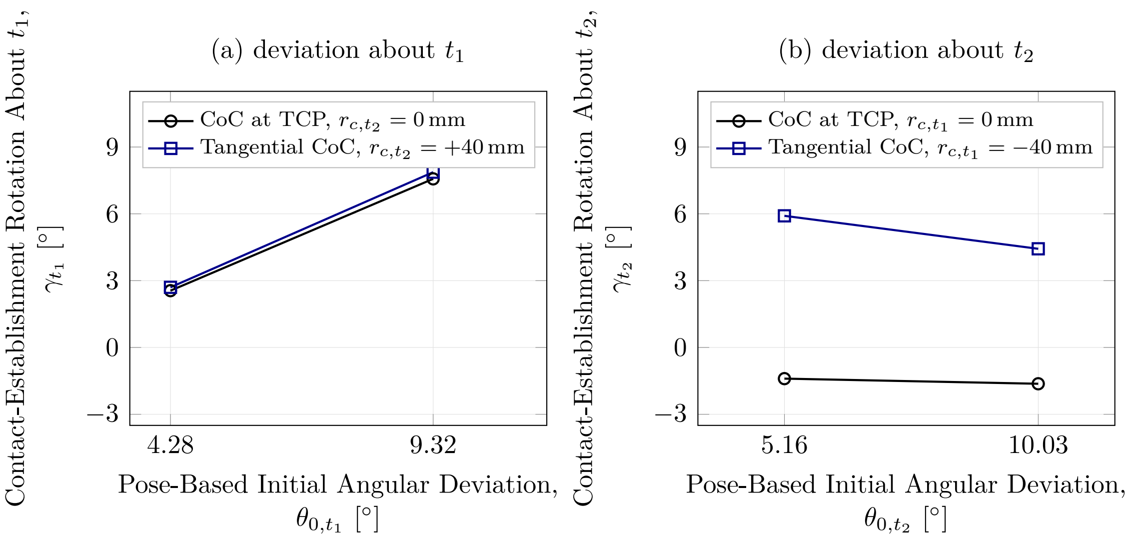

Figure D.1: Contact-establishment rotation by pose-based initial angular deviation.

The response did not scale proportionally with the pose-based initial angular deviation. About t_2, the selected displacement gave +5.91° at 5.16° and +4.43° at 10.03°, as Figure D.1 shows.

D.3 Tool-Axis Displacement

This check separated the distance between the CoC and the TCP from the direction of the displacement. The centre was moved by −40, −20, −10, +10, +20 and +40 mm along the tool axis at a fixed +10° controller input about t_1. The achieved pose-based initial deviations were approximately +9.3°. These six settings gave 18 runs, and the shared zero-displacement condition is counted with the Case-A reference.

Table D.6: Tool-axis displacement response at an initial deviation of approximately +9.3° about t_1.

Measured Response	−40	−20	−10	**0**	+10	+20	+40 mm
γ_{t_1}	+7.26	+7.49	+7.58	+7.57	+7.58	+7.55	+7.60
Standard deviation	0.13	0.01	0.01	0.01	0.01	0.03	0.04

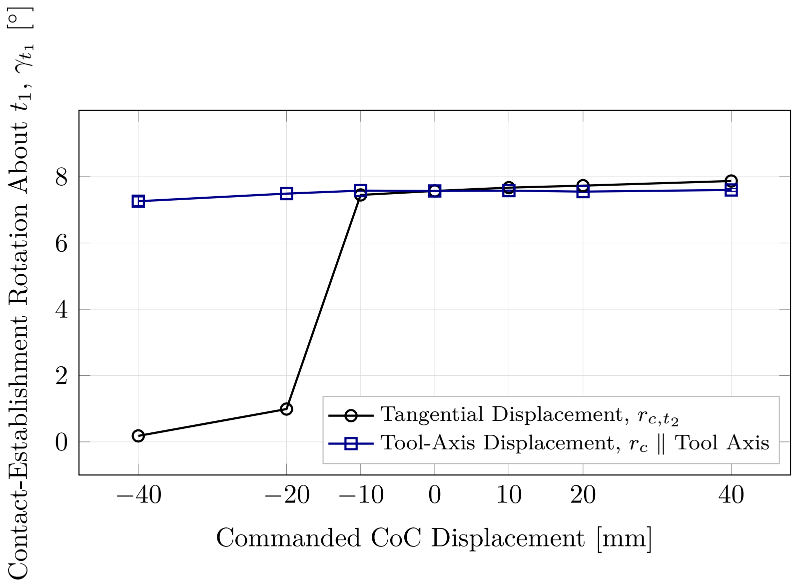

Figure D.2: Tangential and tool-axis centre displacement at the same command.

The signed contact-establishment rotation changed from +7.26 to +7.60° between −40 and +40 mm, a span of 0.34°, against 7.73° for the purely tangential displacement of Case D over the same interval, as Figure D.2 shows. Every tool-axis position was classified as flat.

Only the tangential component of the CoC displacement contributes to the moment generated by the commanded normal force, by Equation (2.48). Let α_{axis} denote the acute inclination of the tool axis relative to the inward surface-normal direction $-n_s$, which is the direction the tool axis takes when the face is flat on the surface. The check used a +10° controller input about t_1. The pose-based initial inclination from $-n_s$ was approximately 9.30°, so a displacement along the tool axis carried a tangential magnitude

$$\|r_{c,t}\| = \|r_c\| \sin \alpha_{\text{axis}}. \tag{D.1}$$

At the largest tested tool-axis displacement, $\|r_c\| = 40\,\text{mm}$. Evaluating the projection at the pose-based inclination, to two decimal places, gives

$$\|r_{c,t}\| = (40\,\text{mm}) \sin(9.30°) = 6.47\,\text{mm}.$$

The 40 mm value is the total configured CoC displacement along the tool axis, and the 6.47 mm value only its projection onto the surface tangent plane. That effective tangential component is considerably smaller than the tangential displacement of Case D, which is consistent with the weak variation measured here. Tangential force components, finite rotation, changing contact geometry and tool-mount compliance were not isolated by this check.

D.4 Intermediate Tangent-Plane Directions

This check examined whether the direction rule of Case D also held between the principal surface tangents. The selected displacement followed Equation (2.51) with $\|r_{c,t}\| = 40\,\text{mm}$. Table D.7 lists the actual vectors used for the four tested directions. The fixed comparison retained the vector selected for the t_1 command.

Table D.7: Commanded directions and the tangential displacements tested against them.

Commanded Rotation Direction	Displacement Setting	Commanded Rotation Components, $[\theta_{t_1}, \theta_{t_2}]$ [°]	CoC Components, $[r_{c,t_1}, r_{c,t_2}, r_{c,n}]$ [mm]
About t_1	selected	$[+10.0, 0.0]$	$[0, +40.0, 0]$
$-45°$ from t_1	selected	$[+7.1, -7.1]$	$[+28.3, +28.3, 0]$
$+45°$ from t_1	selected	$[+7.1, +7.1]$	$[-28.3, +28.3, 0]$
About t_2	selected	$[0.0, +10.0]$	$[-40.0, 0, 0]$
$-45°$ from t_1	fixed at the t_1 selection	$[+7.1, -7.1]$	$[0, +40.0, 0]$
$+45°$ from t_1	fixed at the t_1 selection	$[+7.1, +7.1]$	$[0, +40.0, 0]$
About t_2	fixed at the t_1 selection	$[0.0, +10.0]$	$[0, +40.0, 0]$

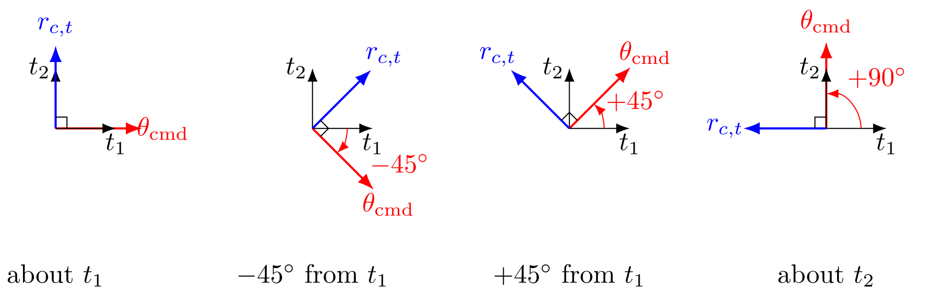

Figure D.3: Selected tangential displacement for each commanded rotation direction.

The principal-axis selected conditions are counted with Case D. The five supporting-check settings were both displacement definitions at each diagonal direction and the fixed displacement for the t_2 command, each repeated three times, giving 15 runs. Every non-zero setting was configured tool-fixed in end-effector coordinates, and the vectors in Table D.7 are the surface-frame components they realise in the flat target orientation.

Table D.8: Direction-rule comparison at $\|r_{c,t}\| = 40\,\text{mm}$.

Commanded Rotation Direction [°]	Selected Displacement $[r_{c,t_1}, r_{c,t_2}]$ [mm]	Selected γ_{t_i} [°]	Fixed at t_1 γ_{t_i} [°]
About t_1	$[0, +40.0]$	+7.87	+7.87
$-45°$ from t_1	$[+28.3, +28.3]$	+4.46	+4.39
$+45°$ from t_1	$[-28.3, +28.3]$	+4.85	+4.80
About t_2	$[-40.0, 0]$	+4.43	−1.61

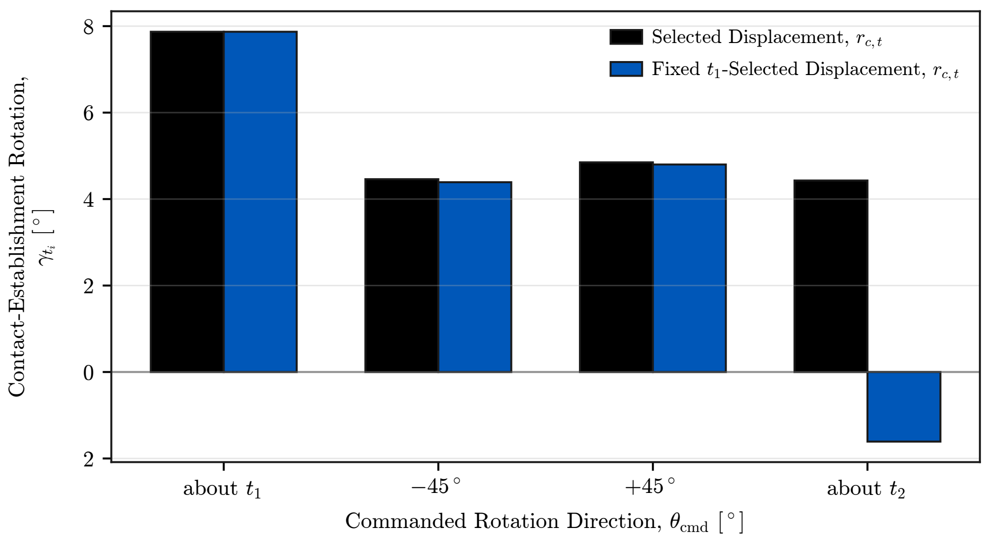

Figure D.4: Response for selected and fixed CoC displacements.

At the two diagonal directions the selected and fixed displacements differed by 0.07° and 0.05°, within the observed scatter, and the fixed displacement retained a substantial projection along the selected direction in both. The clearest separation occurred about t_2, where Table D.8 and Figure D.4 give +4.43° for the selected displacement against −1.61° for the fixed one, a difference of 6.04°. These measurements support the direction rule of Section 2.7.2 over the four tested directions without establishing a single non-zero displacement that suits all of them.

D.5 Conditions Classified as Tilted by TCP Height

The TCP-height classification is retained here as a supporting geometric check and does not enter the response comparisons of Chapter 5. Of the 52 tested parameter settings, 47 satisfied it, including all five Case-A settings; Table D.9 lists the five that did not, with the excess TCP height above the flat-tool height.

Table D.9: Conditions classified as tilted by TCP height.

Condition	Excess height [mm]	γ_{t_1} [°]
Case B, 50 N m/rad at $\theta_{0,t_1} = +9.31°$	3.6	+2.73
Case D, −20 mm at $\theta_{0,t_1} = +9.32°$	5.3	+0.99
Case D, −40 mm at $\theta_{0,t_1} = +9.32°$	6.1	+0.18
Case D, +20 mm at $\theta_{0,t_1} = -9.36°$	5.2	−5.01
Case D, +40 mm at $\theta_{0,t_1} = -9.36°$	9.3	−0.09

The signed contact-establishment rotation does not use the calibrated tool normal. It is obtained from the measured end-effector orientations and resolved in the surface frame, which is why it is the quantity compared between settings.

D.6 Pose-Based Alignment Consistency Check

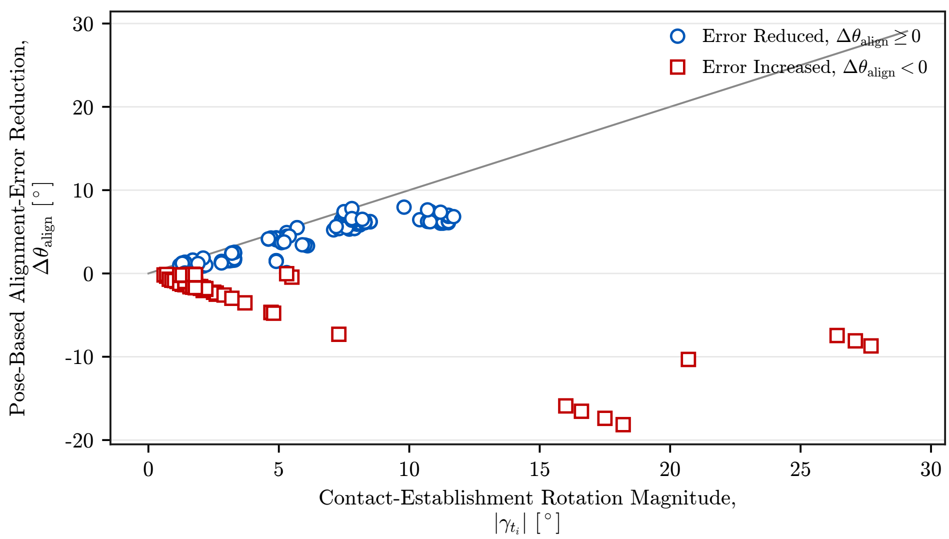

Figure D.5: Pose-based alignment estimate against measured contact-establishment rotation.

The main results use the signed contact-establishment response γ_{t_i}. This appendix compares it with a secondary alignment estimate reconstructed from the calibrated tool normal,

$$\theta_{\text{align}}(t) = \cos^{-1}\left(-n_{\text{Tool},0}(t)^\top n_s\right). \tag{D.2}$$

Here $n_{\text{Tool},0}$ is obtained from Equation (3.1). The angle is unsigned and depends on the assumed fixed relation between the tool and the end effector. Its reduction over contact establishment is

$$\Delta\theta_{\text{align}} = \theta_{\text{align,before}} - \theta_{\text{align,after}}, \tag{D.3}$$

taking $\theta_{\text{align,before}}$ at the start of contact establishment and $\theta_{\text{align,after}}$ from the settled contact-establishment interval. It is used in this appendix alone.

Figure D.5 gives one point per evaluated trial. Each point carries the magnitude of the measured contact-establishment rotation and the signed reduction of the pose-based error. Where the pose-based error decreased, its reduction followed the same scale as the measured rotation for most trials; conditions in which it increased appear below zero.

Their agreement is an internal consistency check rather than an independent measurement of the physical tool orientation, because the pose-based quantity depends on the assumed tool-to-gripper relation while γ_{t_i} is obtained directly from the measured end-effector orientations. The approximately $\pm 2°$ of mounting play was measured unloaded and was not characterised as an uncertainty bound under contact load.